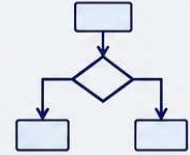


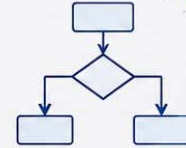
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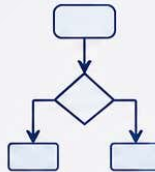
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Unit - I

Introduction to Computer

"A Computer is a machine that turns data into useful information."

1) What is a Computer?

A computer is an electronic device that accepts data as input, processes the data according to given instructions, produces output and stores the result for future use. It can perform millions of calculations in a fraction of a second and can work with high accuracy.

"A computer is a fast, accurate and versatile electronic device used to process data and produce meaningful information."

2) Uses of Computers

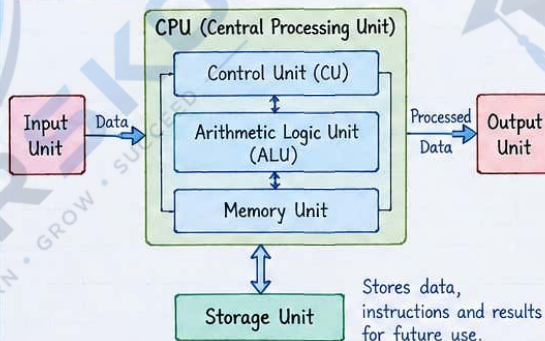
- 1 Education – e-learning, online classes, research.
- 2 Business – accounting, inventory, online banking.
- 3 Science and Research – data analysis, simulations.
- 4 Communication – email, video calling, social media.
- 5 Healthcare – medical records, diagnosis, equipment.
- 6 Government – record keeping, e-governance.
- 7 Entertainment – games, music, movies.
- 8 Industry – automation, design, manufacturing.
- 9 Home – online shopping, bill payment, daily tasks.
- 10 Everywhere – computers are used in almost every field of life.

"Today, computers have become an integral part of our personal, professional and social life."

3) History of Computers

- Early humans used simple tools like abacus for calculations.
- Charles Babbage is known as the "Father of Computer".
- He designed the Analytical Engine (1837) – a mechanical computer.
- Ada Lovelace wrote the first program for this machine and is called the "First Programmer".
- In 1940s, the first electronic computer ENIAC was developed.
- Since then, computers have evolved from large, slow and expensive machines to small, fast and affordable devices.

4) Block Diagram of a Computer



- **Input Unit** – Accepts data and instructions from the user.
- **CPU** – Processes the data using CU and ALU.
- **Control Unit (CU)** – Controls and coordinates all activities.
- **Arithmetic Logic Unit (ALU)** – Performs arithmetic and logical operations.
- **Memory Unit** – Temporarily stores data and instructions.
- **Output Unit** – Displays the processed information.
- **Storage Unit** – Stores data permanently for future use.

5) Characteristics of Computers

- ① **High Speed** – Performs millions of operations per second.
- ② **Accuracy** – Gives correct results with high precision.
- ③ **Diligence** – Does not get tired or bored.
- ④ **Storage Capacity** – Can store huge amount of data.
- ⑤ **Versatility** – Can perform different types of tasks.
- ⑥ **Automation** – Works automatically once instructions are given.
- ⑦ **Reliability** – Produces consistent and reliable results.
- ⑧ **Programmable** – Can be programmed for different tasks.
- ⑨ **No IQ/Emotions** – Works only as per instructions, no feelings.

6) Generations of Computers

Generation	Time Period	Main Features	Examples
First Generation	1940 – 1956	• Used vacuum tubes • Large size, high power consumption • Machine language	ENIAC UNIVAC
Second Generation	1956 – 1963	• Used transistors • Smaller, faster, more reliable • Assembly language	IBM 1401 IBM 1620
Third Generation	1964 – 1971	• Used Integrated Circuits (ICs) • Smaller, more powerful • Operating systems introduced	IBM 360 PDP-8
Fourth Generation	1971 – 1980	• Used microprocessors • Very small, fast, low cost • Personal computers	Intel 4004 Apple II IBM PC
Fifth Generation	1980 – Present	• Uses VLSI, AI and parallel processing • Focus on intelligent systems • Natural language processing	Modern PCs Laptops AI-based systems

"From vacuum tubes to artificial intelligence, computers have come a long way and continue to shape the future."

Unit - I

Introduction to Computer

Uses of Computers

"Computers are used everywhere because they make our work faster, easier and more accurate."

1) Education

- Computers are widely used in schools, colleges and universities for teaching and learning.
- They help in online classes, e-learning, digital libraries and research.
- Students can access study material, attend virtual lectures and take online exams.



Examples : Smart classes, online courses (e.g. SWAYAM), digital libraries, project work, research.

2) Business and Commerce

- Computers are used for accounting, billing, inventory management and financial transactions.
- They help in maintaining records and generating reports.
- Online banking, e-commerce and digital payments have become possible due to computers.



Examples : Tally, online banking, e-commerce websites (Amazon, Flipkart), digital payment (UPI).

3) Science and Research

- Computers are used for data analysis, simulations and experiments.
- Scientists use computers to solve complex problems in fields like physics, chemistry, biology, astronomy and space research.
- They help in modeling, designing and predicting results accurately.



Examples : Weather forecasting, space research (ISRO), medical research, simulations.

4) Communication

- Computers have made communication fast and easy.
- They are used for email, video calling, social media and instant messaging.
- People can connect with anyone, anywhere in the world within seconds.



Examples : Email, WhatsApp, Zoom, Google Meet, social media (Facebook, Instagram).

5) Healthcare

- Computers are used in hospitals and medical fields to maintain patient records, schedule appointments and manage hospital administration.
- They are used in medical imaging (X-ray, MRI, CT Scan), diagnosis and treatment planning.
- They also help in telemedicine and online consultations.



Examples : Patient record management, diagnostic systems, telemedicine, surgery assistance (robotic surgery).

6) Government and Public Services

- Computers are used for maintaining records, planning, monitoring and delivering public services.
- They help in e-governance, online applications, ID systems and digital land records.
- Citizens can access many services online without visiting offices.



Examples : Aadhaar, PAN card, online voting systems, passport services, land records, tax filing.

7) Entertainment

- Computers are used for games, music, movies and multimedia.
- People can watch videos, listen to music, edit photos and play online games.
- Streaming platforms and digital content are possible because of computers.



Examples : YouTube, Netflix, gaming, photo/video editing.

8) Industry and Manufacturing

- Computers are used for automation, design, production planning and quality control.
- They help in controlling machines, robots and assembly lines.
- Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM) are widely used.



Examples : CAD, CAM, robotics, automated production.

9) Home and Personal Use

- Computers are used at home for online shopping, bill payments, managing personal data and entertainment.
- They help in daily tasks like writing documents, budgeting and storing photos.
- They are also used for skill development and online learning from home.



Examples : Online shopping, bill payment, typing, photo storage, learning new skills.

10) Banking and Finance

- Computers are used for managing accounts, transactions and customer records.
- They enable online banking, ATM services and mobile banking.
- They ensure fast, secure and accurate financial operations.



Examples : ATM, net banking, mobile banking, transaction processing, loan management.

11) Transportation

- Computers are used in ticket reservation, traffic management and vehicle control systems.
- They are used in navigation systems (GPS), flight and railway operations.
- They help in managing logistics and supply chains.



Examples : IRCTC, airline booking, GPS navigation, traffic control.

12) Defence and Security

- Computers are used for communication, surveillance, weapon control and data analysis.
- They help in maintaining national security and protecting the country.
- Advanced computers are used in missiles, radar systems and cybersecurity.



Examples : Missile control, satellite monitoring, cybersecurity.

"Computers are used in almost every field of life and have become an essential part of modern society."

Think Digital
Think Better
Think Future

History of Computers

"The history of computers is the story of human intelligence trying to make life easier."

1) What is History of Computers?

- History of computers refers to the development and evolution of computing devices from ancient times to the modern era.
- It shows how humans have always tried to make calculations faster, easier and more accurate.
- Computers have evolved from simple mechanical devices to highly advanced electronic machines used in every field today.

2) Early Need for Computing

- Since ancient times, humans needed to do calculations for trade, astronomy, engineering and daily life.
- As calculations became complex, various devices were invented to help in counting, calculating and storing information.
- This need led to the invention and development of computers.

3) Major Milestones

- From manual calculation to mechanical devices.
- From mechanical to electro-mechanical machines.
- From vacuum tubes to transistors.
- From transistors to integrated circuits.
- From large mainframes to personal computers.
- From personal computers to mobile, cloud and AI-based systems.

4) Definition

"History of computers is the chronological development of devices that can accept data, process it and give meaningful information."

5) Important Stages in the History of Computers

1. Abacus (Ancient Times)



- Used around 3000 BC in China.
- First known tool for counting and basic arithmetic.
- Helped in simple calculations using beads.

2. Napier's Bones (1617)



- Invented by John Napier (Scotland).
- Used for multiplication and division.
- Made calculations faster and easier.

3. Pascaline (1642)



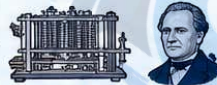
- Invented by Blaise Pascal.
- First mechanical calculator.
- Could perform addition and subtraction.
- Used gears and wheels.

4. Leibniz Calculator (1673)



- Invented by Gottfried Wilhelm Leibniz.
- Could perform addition, subtraction, multiplication and division.
- Used stepped drum mechanism.

5. Analytical Engine (1837)



- Designed by Charles Babbage (Father of Computer).
- A general purpose mechanical computer.
- Used punch cards for input.
- Could perform any calculation.
- Never fully completed.

6. Tabulating Machine (1890)



- Developed by Herman Hollerith.
- Used punch cards to store data.
- Used in the U.S. Census (1890).
- Basis for IBM.

7. Early Electronic Computers (1940s)



- ENIAC (1946) - first general purpose electronic computer.
- Used vacuum tubes.
- Very large in size, high power consumption.
- Performed thousands of calculations per second.

8. First Generation (1940 - 1956)



- Used vacuum tubes.
- Large size, high power consumption, more heat.
- Machine language.
- Examples: ENIAC, UNIVAC.

9. Second Generation (1956 - 1963)



- Used transistors.
- Smaller, faster, more reliable.
- Lower power consumption.
- Assembly language.
- Examples: IBM 1401, IBM 1620.

10. Third Generation (1964 - 1971)



- Used Integrated Circuits (ICs).
- Smaller, more powerful.
- Introduced operating systems.
- Examples: IBM 360, PDP-8.

11. Fourth Generation (1971 - 1980)



- Used microprocessors (VLSI).
- Very small, fast and low cost.
- Personal computers began.
- Examples: Intel 4004, Apple II, IBM PC.

12. Fifth Generation (1980 - Present)



- Based on ULSI, AI, parallel processing.
- More powerful and intelligent systems.
- Focus on natural language processing and AI.
- Examples: Modern PCs, laptops, smartphones.

13. Modern Era (2000 - Present)



- Cloud computing, mobile computing, IoT, AI, big data.
- Computers are smaller, faster and more powerful.
- Used in every field - education, healthcare, business, entertainment, etc.

Key Personalities



Charles Babbage
- Father of Computer



Ada Lovelace
- First Programmer



Herman Hollerith
- Pioneer of Data Processing



Alan Turing
- Father of Modern Computing

"From the Abacus to Artificial Intelligence, the journey of computers shows how human creativity and innovation can change the world."

Past → Present → Smarter Future

Unit - I

Introduction to Computer

Characteristics of Computers

"The characteristics of computers make them powerful, reliable and indispensable in modern life."

1) High Speed

- Computers can perform millions of instructions per second.
- They can process large amounts of data in a fraction of time compared to humans.
- For example, a computer can solve complex calculations in seconds.



Speed
in
Seconds

Example: Mathematical calculations, weather forecasting, data analysis.

2) Accuracy

- Computers give highly accurate results if the input data and instructions are correct.
- They do not make mistakes on their own.
- Accuracy depends on the quality of the input and the program.



Correct
Results
Every Time

Example: Banking transactions, scientific calculations.

3) Diligence

- Computers can work for long hours without getting tired.
- They can perform repetitive tasks again and again with the same speed and accuracy.
- Unlike humans, they do not lose interest or concentration.



No
Fatigue
No Breaks

Example: Repetitive data entry, continuous monitoring systems.

4) Storage Capacity

- Computers can store huge amounts of data and retrieve it whenever required.
- They can store different types of data like text, numbers, images, videos and programs.
- Data can be stored for a long time without loss.



Store
More
Do More

Example: Student records, business data, multimedia files.

5) Versatility

- Computers can perform a wide variety of tasks.
- They can be used in education, business, healthcare, science, entertainment and many other fields.
- They can switch from one task to another quickly.



One Machine
Many Uses

Example: Document editing, gaming, programming, design, research.

6) Automation

- Once a program and instructions are given, the computer works automatically.
- It can complete tasks without human intervention.
- Automation saves time, reduces human effort and increases productivity.



Set
Run
Relax

Example: Automated factories, online systems, data processing.

7) Reliability

- Computers produce consistent and reliable results.
- They can work for long periods without failure.
- They are widely trusted for critical tasks because of their stability and dependability.



Consistent
&
Dependable

Example: Air traffic control systems, hospital management systems.

8) Programmable

- Computers can be programmed for different tasks.
- By changing the software, they can perform new functions.
- They can be used for simple tasks like calculation to complex tasks like artificial intelligence.



Change
Program
Change
Possibilities

Example: Accounting software, graphic design, machine learning.

9) No IQ / No Emotions

- Computers cannot think on their own.
- They have no intelligence, feelings or emotions.
- They work only as per the instructions given by humans.
- They cannot take decisions unless programmed.



No Feelings
Only
Instructions

Example: Cannot judge right or wrong, cannot feel happiness or sadness.

Summary

The main characteristics of computers are:

- High Speed
- Accuracy
- Diligence
- Storage Capacity
- Versatility
- Automation
- Reliability
- Programmable
- No IQ / No Emotions

"These characteristics make computers an essential part of our personal, professional and social life."

Think · Learn · Apply · Grow

Unit - I

Introduction to Computer

"Each generation of computers is a step towards a smarter future."

Generations of Computers

The history of computers can be divided into five generations based on the major technological developments used in their design and construction.

"From vacuum tubes to artificial intelligence, each generation has made computers smaller, faster, more powerful and more useful."

First Generation (1940 - 1956)



ENIAC (1946)

- Used vacuum tubes.
- Very large in size (room-sized).
- High power consumption.
- Generated a lot of heat.
- Very expensive.
- Slow processing speed.
- Used machine language.
- Used for scientific and military calculations.

Examples : ENIAC, UNIVAC, EDSAC, IBM 701.

Second Generation (1956 - 1963)



Transistors

- Used transistors instead of vacuum tubes.
- Smaller, faster and more reliable than first generation.
- Less heat and power consumption.
- Used assembly language.
- More efficient and affordable.
- Used for commercial and business applications.

Examples : IBM 1401, IBM 1620, CDC 1604.

Third Generation (1964 - 1971)



Integrated Circuit (IC)

- Used Integrated Circuits (ICs).
- Smaller, more powerful and more reliable.
- Further reduced size, heat and power consumption.
- Used high-level programming languages (FORTRAN, COBOL).
- Could perform multiple tasks at the same time (multitasking).
- Widely used in universities, businesses and industries.

Examples : IBM 360, PDP-8, CDC 6600.

Fourth Generation (1971 - 1980)



Microprocessor

- Used microprocessors (VLSI) - thousands of transistors on a single chip.
- Much smaller, faster and more powerful.
- Low power consumption.
- Development of personal computers (PCs).
- Used high-level languages like C, C++, Pascal.
- GUIs (Graphical User Interface) began to appear.

Examples : Intel 4004, IBM PC, DEC VAX.

Fifth Generation (1980 - Present)



- Based on ULSI (Ultra Large Scale Integration).
- Extremely small, fast and powerful.
- Development of Artificial Intelligence (AI) and Machine Learning.
- Use of natural language processing, expert systems and robotics.
- Cloud computing, parallel processing and advanced operating systems.
- Widely used in all fields - education, healthcare, industry, entertainment, etc.

Examples : Modern PCs, laptops, smartphones, AI systems (Siri), robotics, supercomputers.

Comparison of Generations

Feature	First	Second	Third	Fourth	Fifth
Time Period	1940-1956	1956-1963	1964-1971	1971-1980	1980-Present
Main Component	Vacuum Tubes	Transistors	Integrated Circuits	Microprocessors	ULSI, AI
Size	Very Large	Large	Smaller	Small	Very Small
Speed	Very Slow	Faster	Fast	Very Fast	Extremely Fast
Power Consumption	Very High	High	Moderate	Low	Very Low
Heat Generation	High	Less	Less	Very Less	Minimal
Programming Language	Machine Language	Assembly Language	High-level (FORTRAN, COBOL)	High-level (C, C++)	AI, Natural Language
Examples	ENIAC, UNIVAC	IBM 1401	IBM 360	Intel 4004	Modern PCs, AI

Advantages of Each Generation

- ✓ First - introduced electronic computing.
- ✓ Second - more reliable and efficient.
- ✓ Third - smaller, faster and multitasking.
- ✓ Fourth - personal computers became possible.
- ✓ Fifth - intelligent systems and advanced capabilities.

Disadvantages

- ✗ First - very large, costly, high heat.
- ✗ Second - still bulky and limited.
- ✗ Third - complex and expensive IC technology.
- ✗ Fourth - security and maintenance issues.
- ✗ Fifth - privacy concerns, job displacement due to AI.

"Generations of computers show how innovation turns imagination into reality."

Unit - I

Introduction to Computer

Block Diagram of a Computer

“A computer works as a system of different units, all connected together to perform input, processing, storage and output operations.”

1) What is a Block Diagram?

- A block diagram is a simple graphical representation of a system.
- It shows the main components of a computer and the flow of data between them.
- It helps to understand how different parts work together.
- Each block represents a specific function.

2) Main Components

A computer system consists of five main units:

- 1) Input Unit
- 2) Central Processing Unit (CPU)
- 3) Memory Unit
- 4) Output Unit
- 5) Storage Unit (Secondary Memory)

5) Input Unit

- Accepts data and instructions from the user.
- Converts them into machine-readable form (binary).
- Examples: Keyboard, mouse, scanner, microphone, barcode reader.

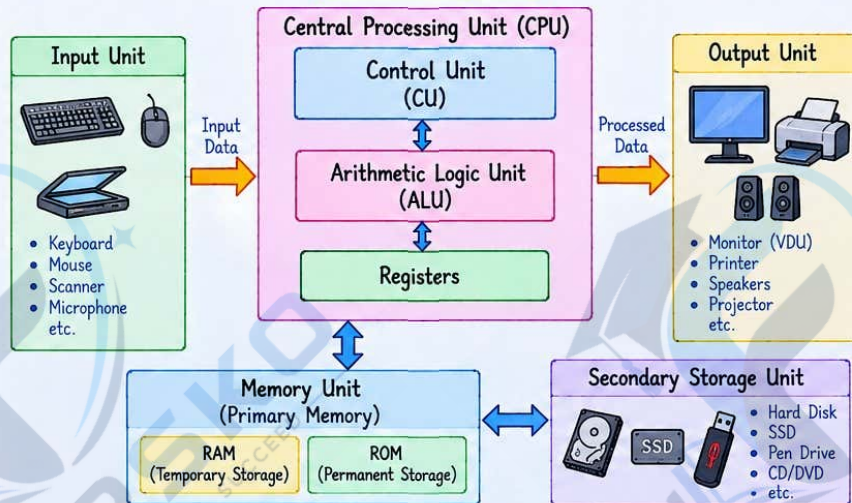
Function: To provide data to the computer.

6) Central Processing Unit (CPU)

- The brain of the computer.
 - Consists of three main parts:
- 1) **Control Unit (CU)** – Controls and coordinates all activities.
 - 2) **Arithmetic Logic Unit (ALU)** – Performs arithmetic and logical operations.
 - 3) **Registers** – Temporary storage inside CPU.

Function: To process data and control the working of all units.

Block Diagram



3) Flow of Data

- Input Unit takes data and converts it into electronic form.
- Data is sent to CPU for processing.
- CPU processes the data and produces meaningful information.
- The processed data is sent to Output Unit.
- Data and instructions are stored in Memory Unit.
- For permanent storage, data is kept in Secondary Storage Unit.

4) Working of a Computer

- 1) **Input** – User gives data through input devices.
- 2) **Processing** – CPU processes the data using ALU and CU.
- 3) **Storage** – Data and instructions are stored in memory.
- 4) **Output** – Results are given through output devices.

8) Secondary Storage Unit

- Stores data permanently.
- Examples: Hard Disk, SSD, Pen Drive, CD/DVD.
- Large storage capacity.
- Data remains even when power is off.

Function: To store data, programs and files for future use.

9) Output Unit

- Displays the processed results to the user.
- Converts machine-readable data into human-readable form.
- Examples: Monitor (VDU), printer, speakers, projector.

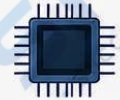
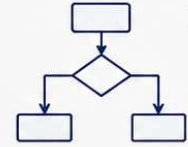
Function: To show the output or results.

“A simple diagram shows a powerful system!”

“All the units of a computer work together to convert data into useful information.”

Input → Process → Output
↓
Storage

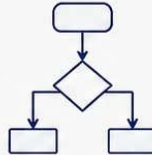
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Unit - II



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Unit - II Memories

Memories in Computer

"Memory is the storage space in a computer that holds data, instructions and information for processing and future use."

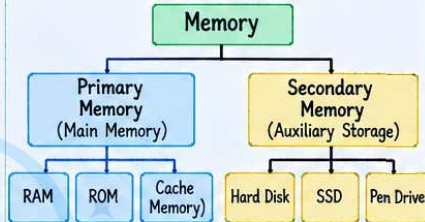
1) What is Memory?

- Memory is a device or part of a computer that is used to store data, instructions and information.
- It stores data temporarily or permanently so that the CPU can use it when required.
- Memory helps the computer to remember data and instructions.
- It is an essential component of a computer system.

Example : Programs, documents, images, videos, operating system files etc.

2) Types of Memory

Memory is mainly divided into two types:



3) Primary Memory (Main Memory)

- Primary memory is directly accessed by the CPU.
- It stores data and instructions temporarily while the computer is running.
- It is fast, has limited storage capacity and volatile in nature (except ROM).
- It is also called Main Memory.

Types :

- 1) RAM (Random Access Memory)
- 2) ROM (Read Only Memory)
- 3) Cache Memory

4) Secondary Memory (Storage Devices)

- Secondary memory is used to store data permanently.
- It is not directly accessed by CPU.
- It has large storage capacity.
- It is non-volatile (data remains even when power is off).
- It is also called Auxiliary Storage Memory.

Types :

- 1) Hard Disk
- 2) SSD (Solid State Drive)
- 3) Pen Drive (USB Flash Drive)

5) RAM (Random Access Memory)

- RAM is a temporary memory.
- It stores data and instructions that are currently being used.
- It is volatile (data is lost when power is off).
- It allows both read and write operations.
- It is fast and helps the CPU to access quickly.



RAM Module

Features :

- Temporary storage
- Volatile
- Fast access
- Read and write both
- Used to store running programs and data

6) ROM (Read Only Memory)

- ROM is a permanent memory.
- It stores important instructions such as the booting program (BIOS).
- It is non-volatile (data remains even when power is off).
- It allows only read operations.
- It is slower than RAM but more reliable.



Types of ROM :

- 1) PROM (Programmable ROM)
- 2) EPROM (Erasable Programmable ROM)
- 3) EEPROM (Electrically Erasable Programmable ROM)

7) Cache Memory

- Cache memory is a small but very fast memory.
- It stores frequently used data and instructions.
- It acts as a bridge between CPU and RAM.
- It is volatile.
- It is faster than RAM and helps in improving the speed of the computer.



Cache Memory

Levels of Cache :

- L1 Cache – inside CPU (fastest, smallest)
- L2 Cache – larger than L1 (slower than L1)
- L3 Cache – largest (shared, slower than L2)

8) Secondary Storage Devices

1) Hard Disk (HDD)



Hard Disk

- Large storage capacity.
- Used to store OS, software, files.
- Slower than SSD.
- Data remains even when power is off.

2) SSD (Solid State Drive)



SSD

- Uses flash memory.
- Faster than hard disk.
- More durable and reliable.
- Large storage capacity.

3) Pen Drive (USB Flash Drive)



Pen Drive

- Portable storage device.
- Uses flash memory.
- Easy to carry and use.
- Used to transfer files between computers.

9) Comparison of Memories

Feature	RAM	ROM	Cache Memory	Hard Disk	SSD	Pen Drive
Type	Primary	Primary	Primary	Secondary	Secondary	Secondary
Nature	Volatile	Non-volatile	Volatile	Non-volatile	Non-volatile	Non-volatile
Storage Capacity	Medium	Low	Very Low	Very High	High	Medium
Speed	Fast	Slow	Very Fast	Slow	Faster	Moderate
Read/Write	Read & Write	Read Only	Read & Write	Read & Write	Read & Write	Read & Write
Use	Running programs	BIOS and permanent data	Frequently used data	Storing large files	High speed storage	Data transfer and storage

10) Importance of Memories

- ✓ Stores data and instructions for processing.
- ✓ Helps in fast and efficient working of computer.
- ✓ Allows multitasking and smooth performance.
- ✓ Different types of memory are used for different purposes.
- ✓ Improves the overall performance and speed of the computer.

"Memory is the backbone of a computer system because it stores data, instructions and information for current and future use."

Unit - II Memories

Primary Memories

(Main Memory of a Computer)

"Primary memory is the working memory of a computer, where data and instructions are stored temporarily (or permanently) for immediate use by the CPU."

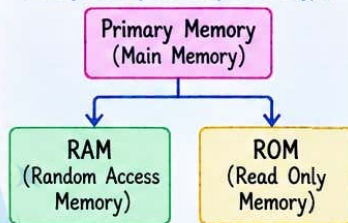
1) What are Primary Memories?

- Primary memory (Main memory) is the memory that is directly accessed by the CPU.
- It stores data and instructions currently being used.
- It is faster than secondary memory.
- It has limited storage capacity.
- It is essential for the execution of programs.
- It is volatile (except ROM).
- It is also called Internal Memory or Main Memory.

Example : RAM, ROM, Cache Memory (also considered as primary memory).

2) Types of Primary Memory

Primary memory is mainly of two types:



3) Random Access Memory (RAM)

- RAM is a temporary memory.
- It stores data and instructions that are currently being used.
- It allows both read and write operations.
- It is volatile (data is lost when power is off).
- It is fast and provides direct access to any memory location.
- It is used to store running programs, operating system and current data.



RAM Module

Features of RAM :

- Volatile
- Read & Write
- Fast access
- Temporary storage

4) Read Only Memory (ROM)

- ROM is a permanent memory.
- It stores important instructions such as the booting program (BIOS).
- It allows only read operations.
- It is non-volatile (data remains even when power is off).
- It is slower than RAM but more reliable.
- It is used to store firmware and system software.



Features of ROM :

- Non-volatile
- Read only
- Permanent storage
- More reliable

5) Comparison: RAM vs ROM

Feature	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Type	Temporary memory	Permanent memory
Volatility	Volatile (data lost when power is off)	Non-volatile (data remains when power is off)
Read/Write	Read and Write	Read only
Speed	Faster	Slower
Storage Capacity	Relatively higher	Relatively lower
Function	Stores current data and instructions	Stores permanent instructions (e.g., BIOS)
Example	DRAM, SRAM	PROM, EPROM, EEPROM

6) Types of RAM

a) DRAM (Dynamic RAM)

- Stores each bit in a capacitor.
- Needs to be refreshed regularly.
- Slower than SRAM.
- Used in main memory of computers.

Example : Used in PCs, laptops, mobile devices.

b) SRAM (Static RAM)

- Stores each bit in a flip-flop (circuit).
- Does not need regular refreshing.
- Faster and more expensive than DRAM.
- Used in cache memory.

Example : Used in cache memory (L1, L2, L3).

7) Types of ROM

a) PROM (Programmable ROM)

- Can be programmed only once.

b) EPROM (Erasable Programmable ROM)

- Can be erased using UV light.
- Can be reprogrammed.

c) EEPROM (Electrically Erasable PROM)

- Can be erased and reprogrammed electrically.
- Used in modern devices.

8) Role of Primary Memory

- Stores data and instructions for immediate processing.
- Helps CPU to access data quickly.
- It is essential for running programs and operating system.
- Acts as a bridge between CPU and secondary storage devices.
- Improves overall performance of the computer.

9) Advantages

- ✓ Faster access to data.
- ✓ Directly accessible by CPU.
- ✓ Essential for program execution.
- ✓ Improves system performance.
- ✓ Different types available for different needs.

10) Disadvantages

- ✗ Limited storage capacity.
- ✗ RAM is volatile (data lost when power is off).
- ✗ More expensive than secondary memory.
- ✗ ROM cannot be easily modified.

Key Point

- Primary Memory = RAM + ROM (+ Cache Memory)
- RAM → Temporary, Read/Write, Volatile
- ROM → Permanent, Read Only, Non-volatile
- Used for fast execution of programs

"Primary memory is the brain's notebook of a computer – it stores what you are working on right now."

Unit - II Memories

RAM (Random Access Memory)

A temporary memory that keeps the computer running

"RAM gives your computer speed, flexibility and the power to handle multiple tasks."



RAM Module

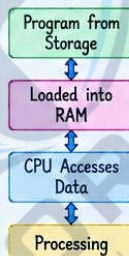
1) Introduction to RAM

- RAM (Random Access Memory) is a type of primary memory in a computer.
- It is a temporary memory that stores data and instructions currently being used by the CPU.
- It allows random access, meaning any memory location can be accessed directly without following a sequence.
- It is volatile, i.e., data is lost when power is turned off.
- RAM helps the computer to run programs faster and smoothly.

Example : When you open a document, game or browser, it is loaded into RAM for quick access.

5) Working of RAM

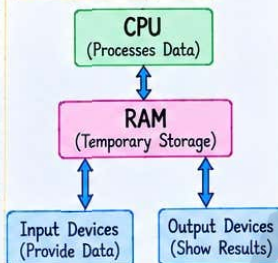
- When a program is executed, it is loaded from secondary storage (like hard disk/SSD) into RAM.
- CPU reads the data and instructions from RAM.
- Data can be written (stored) and read (retrieved) multiple times.
- Any memory location can be accessed directly (random access).
- When power is turned off, the stored data is lost.



2) Full Form and Meaning

- RAM - Random Access Memory
- Random Access - means any memory location can be accessed directly.
- Memory - it is used to store data and instructions temporarily.

3) Block Diagram



4) Characteristics of RAM

- Volatile memory (data lost when power is off).
- Read and write both operations are possible.
- Fast access time compared to secondary memory.
- Random access (any location can be accessed directly).
- Temporary storage for currently running programs.
- Stores data in binary form (0s and 1s).
- Capacity is measured in MB, GB or TB.
- Essential for multitasking and smooth performance.

6) Types of RAM

a) Static RAM (SRAM)

- Uses flip-flop circuit.
- Faster than DRAM.
- More expensive.
- Used in cache memory (L1, L2, L3).
- Does not need to be refreshed regularly.

Example : Cache memory in CPU.

b) Dynamic RAM (DRAM)

- Uses capacitor to store bits.
- Slower than SRAM.
- Cheaper and higher storage capacity.
- Needs to be refreshed regularly.
- Widely used in computers, laptops and mobile devices.

Example : DDR3, DDR4, DDR5.

7) Advantages of RAM

- ✓ Fast access to data.
- ✓ Supports multitasking.
- ✓ Improves overall performance of the computer.
- ✓ Allows read and write operations.
- ✓ Essential for running applications.
- ✓ Available in different capacities (4GB, 8GB, 16GB, etc.).

8) Disadvantages of RAM

- ✗ Volatile (data lost when power is off).
- ✗ More expensive than secondary storage.
- ✗ Limited storage capacity.
- ✗ Requires constant power supply.

9) Applications of RAM

- Running operating systems (Windows, Linux, etc.).
- Running applications (browsers, games, editors, etc.).
- Multitasking (running multiple programs at the same time).
- Used in smartphones, tablets, laptops and desktop computers.
- Used in cache memory (SRAM).

11) Comparison: RAM vs ROM (Quick View)

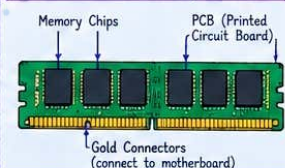
Feature	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Type	Volatile	Non-volatile
Read/Write	Read and Write	Read Only
Speed	Faster	Slower
Use	Temporary storage	Permanent storage
Example	DRAM, SRAM	PROM, EPROM

12) Key Points to Remember

- RAM is a primary memory.
- It is volatile.
- It stores data temporarily.
- It allows random access.
- It is faster than secondary memory.
- Types: SRAM and DRAM.
- Capacity is measured in MB, GB, TB.
- Helps in multitasking and smooth performance.

"More RAM means a smoother, faster and better computing experience."

10) RAM Module (Physical View)



Real Life Example

When you open many tabs in a browser, all of them run in RAM. More RAM allows you to open more tabs without slowing down your computer.

Unit - II Memories

ROM (Read Only Memory)

A permanent memory that stores important instructions

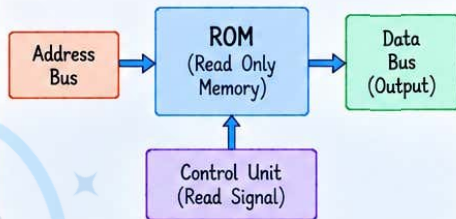
"ROM keeps the essential instructions alive, even when the power is off."

1) Introduction to ROM

- ROM (Read Only Memory) is a type of primary memory.
- It stores data and instructions permanently.
- It is non-volatile, i.e., the data remains even when the power is turned off.
- ROM mainly stores important instructions such as the BIOS (Basic Input Output System).
- It is called "Read Only" because the data stored in it cannot be easily modified.
- ROM is essential for the booting process of a computer.

Example : BIOS, firmware, system startup programs.

2) Block Diagram of ROM



3) Characteristics of ROM

- Non-volatile (data remains even when power is off).
- Stores permanent data and instructions.
- Read only operation (generally cannot be modified).
- Slower than RAM but reliable.
- Used to store firmware and system programs.
- More reliable and less affected by viruses.
- Limited storage capacity compared to hard disk or SSD.
- Data is written during manufacturing or by special devices.

4) Physical Appearance



ROM Chip (DIP Package)



ROM Chip (Surface Mount)

5) Working of ROM

- When the computer is switched on, the CPU sends an address to ROM through the address bus.
- ROM responds by sending the stored data or instructions through the data bus.
- The control unit manages the read operation (using the read signal).
- The data in ROM is hardcoded and cannot be easily changed.
- It is mainly used to store the BIOS, which initializes the hardware and starts the operating system.

Note : ROM does not require power to retain data. It only needs power when data is being read.

6) Types of ROM

a) Mask ROM

- Programmed during manufacturing.
- Cannot be modified.
- Used in mass-produced devices.

Example : Calculators, appliances, embedded systems.

b) PROM (Programmable ROM)

- Can be programmed once by the user using a special device.
- Cannot be erased.
- Used for small scale production or development.

Example : Early microcontrollers, special applications.

c) EPROM (Erasable Programmable ROM)

- Can be erased using ultraviolet (UV) light.
- Can be reprogrammed multiple times.
- Used in development and testing.

Example : Older computers, embedded systems.

d) EEPROM (Electrically Erasable Programmable ROM)

- Can be erased and reprogrammed electrically.
- Does not require UV light.
- More flexible and widely used.

Example : BIOS chips, routers, smart devices.

7) Advantages of ROM

- ✓ Data remains even when power is off (non-volatile).
- ✓ Reliable and secure.
- ✓ Stores essential system instructions.
- ✓ Not affected by viruses easily.
- ✓ Required for booting the computer.
- ✓ Low power consumption.
- ✓ Useful in embedded systems and electronic devices.

8) Disadvantages of ROM

- ✗ Data cannot be easily modified.
- ✗ Limited storage capacity.
- ✗ Slower than RAM.
- ✗ Requires special devices for writing (in case of PROM, EPROM).
- ✗ Not suitable for data that changes frequently.
- ✗ More expensive than RAM (per bit).

9) Comparison: RAM vs ROM

Feature	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Nature	Volatile	Non-volatile
Read/Write	Read and Write	Read Only
Data Retention	Lost when power is off	Retained even when power is off
Speed	Faster	Slower
Storage Capacity	Higher	Lower
Modifiability	Can be modified	Cannot be easily modified
Use	Temporary storage	Permanent instructions

10) Applications of ROM

- Stores BIOS (Basic Input Output System).
- Used in embedded systems.
- Firmware in devices (printers, routers, mobiles, etc.).
- Storing boot instructions.
- Used in calculators, gaming consoles and electronic appliances.

11) Key Points to Remember

- ROM is a type of primary memory.
- It is non-volatile.
- Types: Mask ROM, PROM, EPROM, EEPROM.
- It stores permanent data and instructions.
- It stores permanent data and instructions.
- Mainly used to store BIOS and firmware.
- Cannot be easily modified.

"Without ROM, a computer would not know how to start!"

Unit - II Memories

Cache Memory

A small, fast memory that makes the computer faster

"Cache memory bridges the speed gap between the CPU and main memory."

1) Introduction

- Cache memory is a small, high-speed memory that stores frequently used data and instructions.
- It is located between the CPU and the main memory (RAM).
- It helps to reduce the time taken to access data from the main memory.
- It improves the overall performance of the computer.
- Cache memory is built using Static RAM (SRAM), which is faster and more expensive than the Dynamic RAM (DRAM) used in main memory.

Example : When you open a frequently used program, its data is stored in cache memory for faster access.

5) Levels of Cache Memory

Cache memory is organized in multiple levels to improve performance :

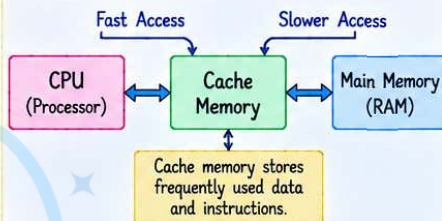
- L1 Cache** - Smallest and fastest. Located inside the CPU.
- L2 Cache** - Larger than L1, slightly slower. Located inside or close to the CPU.
- L3 Cache** - Even larger, slower than L2. Shared among multiple CPU cores.

As we move from L1 → L3, the size increases but the speed decreases.

10) Cache Memory vs Main Memory (RAM)

Feature	Cache Memory	Main Memory (RAM)
Speed	Very fast	Slower
Size	Small	Large
Cost	More expensive	Less expensive
Technology	SRAM	DRAM
Purpose	Stores frequently used data	Stores all data and programs
Location	Between CPU and RAM (inside or near CPU)	External to CPU
Access Time	Very low	Higher
Volatility	Volatile	Volatile

2) Block Diagram



6) Cache Hit and Cache Miss

Cache Hit

- The requested data is found in the cache memory.
- Data is quickly provided to the CPU.
- Access time is very low.
- Improves performance.

Example : Accessing a frequently used variable or instruction.

Cache Miss

- The requested data is not found in the cache memory.
- Data is fetched from main memory (RAM).
- A copy is then stored in cache for future use.
- Takes more time than cache hit.

Example : Accessing a new program or rarely used data.

3) Working of Cache Memory

- When the CPU requests data, it first checks the cache memory.
- If the data is found in the cache (cache hit), it is quickly provided to the CPU.
- If the data is not found (cache miss), it is fetched from the main memory and a copy is stored in the cache for future use.
- Thus, cache memory reduces the average time to access data.

4) Characteristics

- Very high speed compared to main memory.
- Smaller in size than main memory.
- Built using SRAM (Static RAM).
- Stores frequently used data and instructions.
- Reduces memory access time.
- Works automatically (managed by hardware).
- Transparent to the user (user is not aware of its working).
- Expensive compared to main memory.
- Helps in improving system performance and CPU efficiency.

7) Types of Cache

- Instruction Cache**
Stores frequently used instructions.
- Data Cache**
Stores frequently used data.
- Unified Cache**
Stores both instructions and data (used in some modern processors).

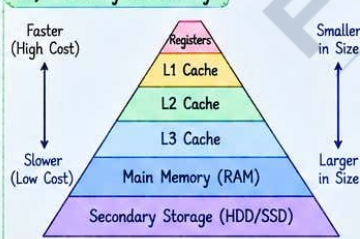
8) Advantages

- ✓ Faster data access.
- ✓ Reduces CPU waiting time.
- ✓ Improves overall system performance.
- ✓ Stores frequently used data and instructions.
- ✓ Bridges the speed gap between CPU and main memory.

9) Disadvantages

- ✗ Smaller storage capacity.
- ✗ More expensive (uses SRAM).
- ✗ Complex hardware design.
- ✗ Cannot store all data (only frequently used data).
- ✗ Data in cache is lost when power is off (since it is volatile).

11) Memory Hierarchy



"Small in size but a big boost in performance - that's Cache Memory!"

12) Applications

- Used in all modern computers and processors.
- Helps in running operating systems, applications and games faster.
- Essential in multitasking and high-performance computing.
- Used in mobile devices, embedded systems, servers, etc.
- Important in real-time systems where speed is critical.

Unit - II Memories

Secondary Storage Devices : Hard Disk

"Hard disk provides high storage capacity for storing data, programs and operating system permanently."

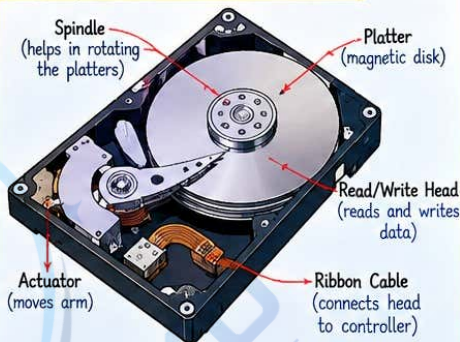
A hard disk is a non-volatile secondary storage device that stores large amount of data permanently.

1) Introduction to Hard Disk

- A hard disk (HDD - Hard Disk Drive) is a non-volatile secondary storage device.
- It uses magnetic storage technology to store data permanently.
- It can store a very large amount of data compared to primary memory (RAM).
- Data remains even when power is turned off.
- It is widely used in desktop computers, laptops, servers and data centers.
- It provides high storage capacity at a lower cost per GB.
- It is also known as a magnetic disk.

Example : Used to store operating system, software, documents, photos, videos, etc.

2) Physical Structure of Hard Disk



Hard Disk Drive (Internal View)

3) Main Components and their Functions

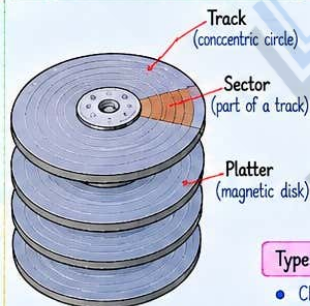
Component	Function
Platters	Circular disks coated with magnetic material to store data.
Spindle	Rotates the platters at high speed (5400, 7200 or 10000 RPM).
Read/Write Head	Reads data from and writes data onto the platter surface.
Actuator Arm	Moves the read/write head to the required track (inward or outward).
Actuator	Electromagnetic device that moves the arm accurately.
Controller	Manages the read/write operations and communicates with the computer.
Ribbon Cable	Connects read/write head to the controller.
Casing	A sealed enclosure to protect the disks from dust and damage.

4) Working of Hard Disk

- Platters rotate at a high speed (e.g., 5400 or 7200 RPM).
- Data is stored on the surface of the platters using magnetic technique.
- The read/write head moves to the required track and sector.
- To write data, the head changes the magnetic polarity on the surface.
- To read data, the head detects the magnetic polarity.
- Data is organized in tracks and sectors.
- The controller manages the data transfer between the hard disk and the computer.

Note : Hard disk uses magnetic storage technology.

5) Internal Organization



Multiple Platters

- A platter is divided into concentric circles called tracks.
- Each track is further divided into small parts called sectors.
- Data is stored in the form of magnetic bits (0 and 1).
- The combination of track number, surface (head) number, and sector number is called an address.

Types of Addressing :

- CHS (Cylinder, Head, Sector)
- LBA (Logical Block Addressing)

6) Key Terms

Term	Meaning
Track	A circular path on a platter where data is stored.
Sector	A small part of a track; minimum storage unit (usually 512 bytes or 4 KB).
Cylinder	A set of tracks at the same position on all platters.
Cluster	A group of sectors (used by operating system).
RPM	Rotations per minute (speed of platters).
Seek Time	Time taken to move the head to the required track.
Latency Time	Time taken for the desired sector to rotate under the head.
Access Time	Total time = Seek time + Latency time.

7) Characteristics of Hard Disk

- Non-volatile (data remains even when power is off).
- Very large storage capacity (GB to TB).
- Faster than other secondary storage devices like CD/DVD and Pen Drive.
- Slower than primary memory (RAM).
- Low cost per GB.
- Uses magnetic storage technology.
- High durability and reliability.
- Data can be read and written multiple times.
- Used for permanent storage of operating system, software and user data.

8) Advantages

- ✓ Large storage capacity.
- ✓ Low cost per GB.
- ✓ Non-volatile (data is permanent).
- ✓ Can read and write data multiple times.
- ✓ Reliable and durable.
- ✓ Widely used in all types of computers.

9) Disadvantages

- ✗ Slower than RAM.
- ✗ Moving parts can be damaged (mechanical failure).
- ✗ More power consumption than SSD.
- ✗ Sensitive to physical shock.
- ✗ Higher seek and latency time.
- ✗ Generates heat and noise.

10) Applications of Hard Disk

- Storing operating system (Windows, Linux, etc.).
- Storing software, games and applications.
- Storing documents, images, videos, etc.
- Used in desktop computers, laptops, servers and data centers.
- Used for backup and archival storage.
- Commonly used in personal and professional computing.

Secondary Storage Devices : SSD

A Solid State Drive (SSD) is a non-volatile secondary storage device that uses flash memory to store data.

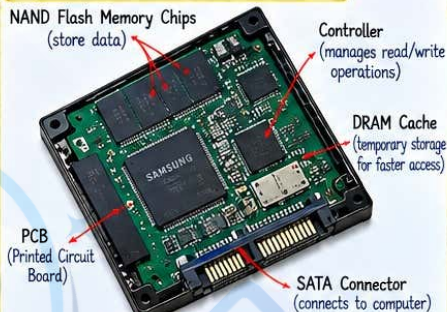
"SSD brings speed, reliability and a better computing experience."

1) Introduction to SSD

- SSD (Solid State Drive) is a non-volatile secondary storage device.
- It uses flash memory chips to store data instead of magnetic disks (as in HDD).
- It stores data electrically, with no moving parts.
- Data remains even when the power is turned off.
- It provides very high speed, low power consumption, and high reliability.
- Widely used in laptops, desktops, servers and modern devices.

Example : Used to store operating system, applications, games, documents, videos, etc.

2) Internal Structure of SSD



Inside a Solid State Drive (SSD)

3) How SSD Works?

- Data is stored in NAND flash memory cells (as electrical charges).
- When data is written, the controller converts it into electrical signals and stores it in memory cells.
- When data is read, the controller retrieves the data from the memory cells and sends it to the computer.
- Uses advanced techniques like wear leveling, error correction and garbage collection to improve performance and lifespan.
- Since there are no moving parts, access time is very low and it is faster than HDD.

4) Types of SSD (Based on Interface)

1) SATA SSD

- Uses SATA interface.
- Slower than NVMe SSD.
- Common in older systems.
- Speed: up to 550 MB/s.

2) NVMe SSD

- Uses PCIe (NVMe) interface.
- Much faster than SATA SSD.
- Used in modern laptops and desktops.
- Speed: 3000 MB/s to 7000 MB/s.

3) M.2 SSD

- Compact form factor.
- Can use SATA or NVMe interface.
- Widely used in modern laptops.

4) U.2 SSD

- Uses PCIe interface (similar to NVMe).
- Common in enterprise systems and servers.

5) SSD vs HDD (Comparison)

Feature	SSD	HDD
Storage Technology	Flash memory (no moving parts)	Magnetic disk (moving parts)
Speed	Very fast	Slower
Access Time	Very low (microseconds)	Higher (milliseconds)
Storage Capacity	Up to several TB	Up to several TB (higher)
Cost per GB	Higher	Lower
Power Consumption	Lower	Higher
Noise	Silent	Produces noise
Durability	More reliable (shock resistant)	Less reliable (mechanical failure)
Heat Generation	Less heat	More heat
Life Span	Longer	Shorter
Use	Modern systems, high performance	Large storage, backup

6) Characteristics of SSD

- Non-volatile (data remains even when power is off).
- No moving parts.
- Very fast read and write speed.
- Low access time (microseconds).
- Low power consumption.
- Silent operation (no noise).
- More reliable and shock resistant.
- Uses flash memory technology.
- Supports advanced features like TRIM, wear leveling and error correction.
- Higher cost per GB compared to HDD.
- Limited number of write cycles (but managed by the controller).

7) Advantages of SSD

- ✓ Faster boot time and application loading.
- ✓ High read/write speed.
- ✓ Low power consumption (better battery life in laptops).
- ✓ Silent operation (no noise).
- ✓ More reliable and durable.
- ✓ Shock and vibration resistant.
- ✓ Compact and lightweight.
- ✓ Generates less heat.
- ✓ Improves overall system performance.
- ✓ Ideal for modern operating systems and gaming.

8) Disadvantages of SSD

- ✗ Higher cost per GB.
- ✗ Limited number of write/erase cycles (though sufficient for normal use).
- ✗ Data recovery is difficult if damaged.
- ✗ Storage capacity is generally lower than HDD (for the same price).
- ✗ Performance may decrease when almost full (due to less free space).
- ✗ Not as cost-effective for large-scale storage or long-term archiving.

9) Applications of SSD

- Storing operating system (Windows, Linux, etc.).
- Running applications and games.
- Used in laptops, desktops, servers and data centers.
- Used in high-performance computing, video editing and gaming.
- Storing frequently used files and data.
- Used in mobile devices, embedded systems and enterprise systems.

10) Key Terms

Term	Meaning
NAND Flash	Type of non-volatile memory used in SSD.
Controller	Manages read/write operations.
TRIM	Helps maintain performance by deleting unused data.
Wear Leveling	Distributes write operations evenly to increase lifespan.
NVMe	A high-speed interface using PCIe.

"Choose SSD for speed, reliability and a smarter future!"

Faster Storage
Brighter Possibilities

Unit - II Memories

Secondary Storage Devices : Pen Drives

"Big in size,
Small in utility
- Pen Drive!"

A pen drive (USB Flash Drive) is a portable secondary storage device that uses flash memory to store data.

1) Introduction to Pen Drives

- A pen drive (also called USB flash drive) is a portable, non-volatile secondary storage device.
- It uses flash memory to store data electronically.
- It is small in size, lightweight and easy to carry.
- Data remains stored even when it is disconnected from the computer (non-volatile).
- It is widely used for storing, transferring and backing up files.

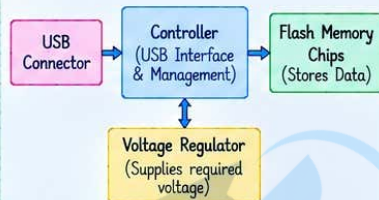
Example : Used to store documents, photos, videos, software, etc.

2) Physical Appearance



Pen Drive (USB Flash Drive)

3) Internal Structure



Block Diagram of a Pen Drive

4) How Pen Drive Works?

- When the pen drive is plugged into a USB port, the computer detects it as a storage device.
- The controller manages the read and write operations between the computer and flash memory chips.
- Data is stored in flash memory cells electronically (no moving parts).
- When files are copied, the data is divided into small blocks and stored in memory cells.
- Data can be read, written and erased electrically.
- Since it uses flash memory, it is faster, more durable and consumes less power than HDD.

5) Types of Pen Drives

- USB 2.0 Pen Drive**
 - Older version.
 - Speed: up to 480 Mbps.
 - Suitable for basic use.
- USB 3.0 / 3.1 / 3.2 Pen Drive**
 - Faster than USB 2.0.
 - Speed: up to 5 Gbps (USB 3.0), 10 Gbps (USB 3.1), 20 Gbps (USB 3.2).
 - Widely used in modern devices.
- OTG Pen Drive**
 - Has dual connectors (USB + Micro USB or USB Type-C).
 - Can be used with smartphones and tablets.
 - Useful for transferring data directly between mobile and computer.

4) Encrypted Pen Drive

- Provides password protection.
- Used for secure and confidential data.

6) Characteristics of Pen Drive

- Non-volatile (data remains even when power is off).
- Portable and lightweight.
- Small in size but available in large capacities (GB to TB).
- Fast read and write speed (especially USB 3.0 and above).
- Plug and play (no need of additional software).
- Uses flash memory (no moving parts).
- Low power consumption.
- Durable and shock resistant.
- Can be rewritten multiple times (limited write cycles).
- Compatible with almost all operating systems (Windows, Linux, Mac, etc.).

7) Advantages of Pen Drive

- ✓ Portable and easy to carry.
- ✓ Large storage capacity.
- ✓ Faster data transfer (especially USB 3.0+).
- ✓ Low power consumption.
- ✓ No moving parts (more durable).
- ✓ Plug and play (easy to use).
- ✓ Can be used on multiple devices.
- ✓ Useful for backup and file transfer.
- ✓ Available in different storage capacities (8GB, 16GB, 32GB, 64GB, 128GB, 256GB, 512GB, 1TB, etc.).
- ✓ Cost-effective compared to CDs/DVDs.

8) Disadvantages of Pen Drive

- ✗ Can be lost easily due to small size.
- ✗ Limited number of write/erase cycles (though sufficient for normal use).
- ✗ More prone to physical damage, water and heat.
- ✗ Can be infected by viruses or malware.
- ✗ Data can be accidentally deleted or formatted.
- ✗ Lower storage capacity compared to hard disks (HDD).
- ✗ May become slow if used for a long time or with low-quality devices.

9) Comparison: Pen Drive vs Other Storage Devices

Feature	Pen Drive	HDD	SSD
Storage Technology	Flash Memory	Magnetic Disk	Flash Memory
Moving Parts	No	Yes	No
Speed	Fast (USB 2.0/3.0)	Slower	Very Fast
Storage Capacity	8GB - 1TB	500GB - 10TB	256GB - 4TB
Portability	High	Low	Medium
Durability	High	Lower (mechanical)	High
Cost per GB	Moderate	Low	Higher
Use	File transfer, backup	Large storage	High performance storage

9) Applications of Pen Drive

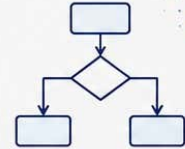
- Storing and transferring documents, photos, videos, music, etc.
- Backup and data storage.
- Installing operating systems (bootable USB).
- Used in schools, colleges, offices and industries.
- Sharing files between computers.
- Used in presentations and project work.
- Carrying portable software and applications.
- Useful for data recovery and system repair.

11) Key Points to Remember

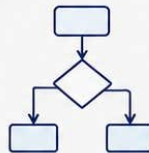
- Pen drive is a secondary storage device.
- It uses flash memory.
- It is non-volatile.
- It is portable, small and lightweight.
- Data can be read, written and erased electronically.
- No moving parts.
- Available in different storage capacities.
- Common interfaces: USB 2.0, 3.0, 3.1, 3.2, Type-C, OTG.
- Widely used for data transfer, backup and storage.

"Carry your data anywhere, anytime - Pen Drive makes it simple!"

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Unit - II

I/O Devices

Input / Output (I/O) Devices

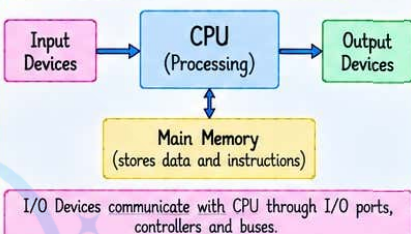
"I/O devices make computers useful by connecting people, data and the real world."

I/O devices are the interface between the computer and the outside world. They allow users to input data and receive output.

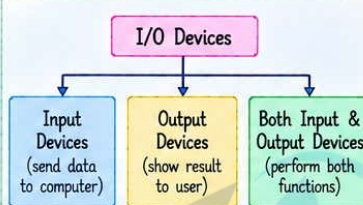
1) Introduction to I/O Devices

- I/O devices are hardware components used to give input to a computer and get output from it.
- Input devices send data and instructions to the computer.
- Output devices present the processed information to the user.
- Some devices can perform both input and output (e.g., touch screen).
- I/O devices make the computer user-friendly and interactive.

2) Block Diagram of I/O System



3) Classification of I/O Devices



4) Characteristics of I/O Devices

- Provide interface between user and computer.
- Accept data in various forms (text, audio, video, etc.).
- Provide output in user-friendly format.
- Work at different speeds and capabilities.
- Can be connected through ports (USB, HDMI, etc.).
- May be dedicated (single function) or multifunctional.
- Require device drivers to communicate with the operating system.

5) Input Devices

Input devices are used to enter data and instructions into the computer.

Keyboard - Enters text, numbers and commands. - Most commonly used input device.	Mouse - Pointing device. - Used to select, click and drag.	Scanner - Converts paper documents to digital form. - Used for images, text, photos, etc.
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Microphone - Input device for audio data. - Used for voice recording, speech recognition, etc.	Webcam - Captures video and images. - Used for video conferencing, etc.	Joystick - Used for gaming and graphics input. - Controls movement in games and simulations.
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Light Pen - Used to select objects on screen. - Commonly used in CAD and gaming.	Touch Screen - Accepts input by finger touch. - Used in smartphones, tablets, ATMs, etc.	Barcode Reader - Reads barcode information. - Used in shops, libraries, etc.
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6) Output Devices

Output devices are used to present the processed data and information to the user.

Monitor - Displays text, images, videos, etc. - Types: CRT, LCD, LED, OLED.	Printer - Produces hard copy on paper. - Types: Inkjet, Laser, Dot Matrix, etc.	Speakers - Produces audio output. - Used for music, videos, alerts, etc.
Headphones - Personal audio output device. - Used for private listening.	Projector - Projects output on a large screen. - Used in presentations, classrooms, etc.	Plotter - Produces high-quality graphics on large paper. - Used in engineering and architectural work.

9) Importance of I/O Devices

- Enable user interaction with the computer.
- Help in data input, processing and result display.
- Used in education, business, healthcare, entertainment, etc.
- Make the computer more versatile and useful.

7) I/O Devices (Both Input & Output)

These devices can perform both input and output functions.

Touch Screen - Displays output and accepts input. - Used in smartphones, ATMs, kiosks, etc.	Modem - Sends and receives data over a network. - Used for internet communication.
Network Interface Card (NIC) - Sends and receives data over a network. - Provides both input and output functionality.	Storage Devices (e.g., External HDD, SSD, Pen Drive) - Reads data (input) and writes data (output). - Used for data transfer and backup.

10) Examples in Real Life

- Typing a document (Keyboard - Input, Monitor - Output)
- Listening to music (Speakers/Headphones - Output)
- Scanning a document (Scanner - Input)
- Video call (Webcam - Input, Monitor - Output)
- Using a pen drive (Input & Output)
- Printing a report (Printer - Output)

8) Connection Interfaces

- I/O devices are connected to the computer through ports and interfaces.
- Common interfaces:
 - 1) USB (Universal Serial Bus) - used for keyboard, mouse, pen drive, etc.
 - 2) HDMI - used for monitor, projector, etc.
 - 3) Audio Jack - used for microphone, headphones, speakers.
 - 4) Ethernet Port - used for network devices.
 - 5) Bluetooth - used for wireless devices (keyboard, mouse, headphones, etc.).
 - 6) Serial / Parallel Ports - older technology.

11) Key Points to Remember

- Input devices send data to the computer.
- Output devices present data to the user.
- Some devices can perform both input and output.
- I/O devices use different ports and interfaces.
- Device drivers are required for communication.
- I/O devices make computers user-friendly and interactive.

"Better I/O devices, better interaction, brighter possibilities!"

Input Device - Keyboard

"A simple device that gives you the power to communicate with a computer."

A keyboard is a primary input device used to enter text, numbers, commands and control signals into a computer.

1) Introduction

- A keyboard is an input device that consists of a set of keys used to enter data into a computer.
- It allows the user to input text, numbers, symbols and special commands.
- It is one of the most commonly used input devices.
- Keystrokes are converted into electronic signals and sent to the CPU for processing.
- Keyboards are essential for communication between the user and the computer.

Example : Typing a document, writing code, entering data in a form, playing games, etc.

2) Keyboard Layout (QWERTY)



3) Types of Keys

Type of Key	Description
Alphanumeric Keys	A-Z, 0-9, used to enter text and numbers.
Function Keys	F1 to F12, perform special functions.
Control Keys	Ctrl, Alt, Shift, used with other keys for shortcut commands.
Navigation Keys	Arrow keys, Home, End, Page Up, Page Down, used for cursor movement.
Numeric Keypad	Keys 0-9, arithmetic operators, used for fast number entry (on right side).
Special Keys	Enter, Backspace, Tab, Esc, Spacebar, etc.

4) How a Keyboard Works?

- When a key is pressed, a mechanical or electronic switch is activated.
- The key sends a unique scan code to the keyboard controller.
- The controller converts it into a digital signal.
- This signal is sent to the computer through a wired (USB/PS/2) or wireless (Bluetooth) connection.
- The operating system translates the scan code into the corresponding character or command.
- The character is then displayed on the screen or processed by the application.

5) Types of Keyboards

- Based on Connection**
 - Wired Keyboard (USB, PS/2)
 - Wireless Keyboard (Bluetooth, RF)
- Based on Design**
 - Standard Keyboard
 - Ergonomic Keyboard (curved design for comfort)
 - Multimedia Keyboard (extra keys for media control)
 - Gaming Keyboard (high speed, programmable keys)
- Based on Key Mechanism**
 - Membrane Keyboard (quiet, low cost)
 - Mechanical Keyboard (durable, better feedback)
 - Scissor Switch Keyboard (used in laptops)
- Based on Layout**
 - QWERTY (most common)
 - Others: AZERTY, DVORAK, etc.

6) Advantages

- Easy to use.
- Fast and accurate data entry.
- Widely available and affordable.
- Compatible with almost all computers.
- Essential for daily tasks (typing, coding, etc.).
- Different types available for different needs.
- Multimedia and shortcut keys improve productivity.

7) Disadvantages

- Requires physical contact (not touch-based).
- Can be damaged by liquid spills or dust.
- May cause health issues like wrist strain with prolonged use.
- Limited mobility in wired keyboards.
- Occupies desk space.
- Keys may wear out over time.
- Not suitable for users with physical disabilities (without assistive features).

8) Uses / Applications

- Typing documents, letters and reports.
- Programming and coding.
- Data entry in offices, schools and banks.
- Controlling computer ~~operation~~ multimedia.
- Used in gaming and multimedia.
- Essential in almost every field (education, business, healthcare, etc.).

9) Key Points to Remember

- A keyboard is an input device.
- It converts keystrokes into digital signals.
- QWERTY is the most common keyboard layout.
- Different types of keyboards exist for different needs.
- It is widely used for typing, data entry and control commands.
- Proper use and maintenance increase its lifespan.

Small Keys... Big Possibilities... * Keyboard connects your ideas to the digital world!"

"Type Create Explore With a Keyboard!"

Input Device - Mouse

"Small device...
Big control...
The Mouse makes
computing easy!"

A mouse is a pointing input device used to move the cursor on the screen and select, open or manipulate items.

1) Introduction

- A mouse is a pointing input device used to control the movement of a cursor on the screen.
- It allows the user to select, open, move and manipulate items.
- It was invented by Douglas Engelbart in 1964.
- The mouse is one of the most widely used input devices in modern computers.
- It makes the computer user-friendly and easy to operate, especially with Graphical User Interfaces (GUIs).

Example : Selecting icons, opening files, clicking buttons, dragging and dropping objects, etc.

2) Parts of a Mouse



3) How a Mouse Works?

- The user moves the mouse on a flat surface.
- Sensors inside the mouse detect the movement (directional changes).
- The movement is converted into electronic signals.
- These signals are sent to the computer through a cable (USB/PS2) or wirelessly (Bluetooth/RF).
- The operating system processes the signals and moves the pointer (cursor) on the screen accordingly.
- When a button is clicked, a signal is sent to perform a specific action (e.g., open, select).
- The scroll wheel generates signals to scroll the content up or down.

4) Types of Mouse

1) Mechanical Mouse

- Uses a rubber ball to detect movement.
- Older technology.



2) Optical Mouse

- Uses an LED and light sensor to detect movement.
- No moving parts.



3) Laser Mouse

- Uses a laser for high precision.
- Works on more surfaces.



4) Wireless Mouse

- Uses Bluetooth or RF for connectivity.
- No physical cable.



5) Functions of a Mouse

- Move the cursor/pointer on the screen.
- Select and open files, folders and applications.
- Click, double-click and right-click to perform different actions.
- Drag and drop objects.
- Scroll through documents and web pages.
- Used in drawing, designing and gaming.
- Helps in navigating Graphical User Interfaces (GUIs).

6) Common Mouse Actions

Action	How to Do	Use
Single Click	Press left button once	Select an item
Double Click	Press left button twice quickly	Open a file or application
Right Click	Press right button once	Show context menu
Scroll	Rotate scroll wheel	Scroll up or down
Drag and Drop	Hold left button and move	Move objects
Hover	Move pointer over an item	Show information (tooltip)

7) Advantages

- ✓ Easy to use and user-friendly.
- ✓ Makes navigation simple and fast.
- ✓ Widely available and affordable.
- ✓ Improves productivity.
- ✓ Essential for modern operating systems.
- ✓ Available in wired and wireless options.
- ✓ Useful in various fields like education, business, gaming and design.

8) Disadvantages

- ✗ Requires a flat surface to work properly.
- ✗ Mechanical mouse can get dirty (due to moving parts).
- ✗ May not work well on reflective or transparent surfaces (optical mouse).
- ✗ Can cause hand or wrist strain with prolonged use.
- ✗ Needs batteries in wireless mouse.
- ✗ Limited functionality compared to some advanced input devices (e.g., touch screen, stylus).

9) Mouse Connectors

USB Mouse

- Most common.
- Plug and play.
- Supported by almost all computers.



PS/2 Mouse

- Older type connector (usually purple).
- Used in older computers.



10) Applications of Mouse

- Used in operating systems (Windows, Linux, Mac, etc.).
- Web browsing.
- Graphic designing and photo editing.
- Gaming.
- Education and e-learning.
- Office work (word processing, spreadsheets, presentations).
- Used in ATMs, kiosks, control systems, etc.

11) Key Points to Remember

- A mouse is a pointing input device.
- Invented by Douglas Engelbart in 1964.
- Common types: Mechanical, Optical, Laser, Wireless.
- Main functions: point, click, drag, scroll.
- Makes computer operation easier and faster.
- Proper use and maintenance increase its life span.

"Move
Click
Create
Explore
All with a Mouse!"



"A small move... for a big digital world!"

Input Device - Scanner

"Scan Today
Store Forever
- Turn Paper into
Possibilities!"

A scanner is an input device that converts physical documents, images or objects into digital form.

1) Introduction

- A scanner is an input device that captures text, images or objects from the physical world and converts them into digital form.
- It creates a digital copy (scan) of a document or image, which can be stored, edited or shared on a computer.
- Scanners use light and sensors to read the content of the original document.
- They are widely used in home, office, schools, hospitals, libraries and industries.
- Scanners help in reducing paper work and make document management easy.

Example : Scanning certificates, photos, forms, books, ID cards, etc.

2) Working of a Scanner

- The document is placed on the scanner's glass surface (or fed through an automatic feeder).
- A light source (usually a lamp or LED) illuminates the document.
- The reflected light from the document is captured by sensors (CCD or CIS).
- The analog image is converted into digital data using an Analog to Digital Converter (ADC).
- The scanned image is sent to the computer and saved in formats like JPG, PNG or PDF.

3) Block Diagram of a Scanner



4) Main Components of a Scanner

Component	Function
Glass Surface	Document is placed on it.
Light Source (Lamp/LED)	Illuminates the document.
Mirrors and Lenses	Reflect and focus the light on the sensor.
Image Sensor (CCD/CIS)	Captures the reflected light and converts it into an electrical signal.
Analog to Digital Converter (ADC)	Converts analog signal into digital data.
Control Circuitry	Controls the scanning process.
Interface	Connects the scanner to the computer (USB, etc.).

5) Types of Scanners

1) Flatbed Scanner

- Most common type.
- Document is placed on a flat glass surface.
- Suitable for photos, documents, books, etc.



2) Sheet-fed Scanner

- Uses an automatic feeder to pull pages.
- Faster than flatbed.
- Used for large number of documents.



3) Handheld Scanner

- Small and portable.
- Moved manually over the document.
- Used for small documents and on-the-go scanning.



4) Drum Scanner

- High quality and high resolution.
- Used in professional publishing and industries.
- Expensive.



6) Common Specifications

Specification	Description
Resolution (DPI)	Dots per inch. Higher DPI gives better quality (e.g., 300, 600, 1200 DPI).
Color Depth	Number of bits used to represent color (e.g., 24-bit, 48-bit).
Scan Speed	Time taken to scan a page (measured in pages per minute - ppm).
Scan Area	Maximum document size (e.g., A4, A3).
Interface	USB, Wi-Fi, etc.
Output Formats	JPG, PNG, PDF, TIFF, etc.

7) Advantages

- ✓ Converts paper documents to digital form.
- ✓ Easy to use.
- ✓ Saves storage space.
- ✓ Helps in editing, sharing and printing documents.
- ✓ Reduces paper work and keeps documents safe.
- ✓ Available in different types and price ranges.
- ✓ Useful in offices, schools, hospitals, libraries, etc.

8) Disadvantages

- ✗ Good quality scanners can be expensive.
- ✗ Some scanners are large and require more space.
- ✗ High resolution scanning increases file size.
- ✗ Cannot scan very thick or damaged documents.
- ✗ Color scanning may require more storage.
- ✗ Regular cleaning and maintenance is needed.

9) Applications of Scanner

- Scanning documents (reports, forms, certificates, etc.).
- Scanning photos and images.
- Used in offices for record keeping.
- Used in schools and colleges for digitizing study material.
- Used in hospitals for patient records.
- Used in libraries for digitizing books.
- Useful in banking, government offices and industries.

10) Key Points to Remember

- A scanner is an input device.
- It converts physical documents into digital form.
- Common types: Flatbed, Sheet-fed, Handheld, Drum.
- Resolution is measured in DPI.
- Uses light and sensors (CCD or CIS).
- Output can be saved in formats like JPG, PNG, PDF.
- Widely used in homes, offices, schools, hospitals, etc.
- Helps in creating a paperless environment.

"Scan
Save
Share
Simplify!"



From Paper to Pixels
A Smarter Tomorrow!"

Output Device - Touch Screen

A touch screen is an output device that displays information on the screen and also allows the user to give input by touch.

"See it...
Touch it...
Do it!
Touch Screen Makes
Technology Closer to You!"

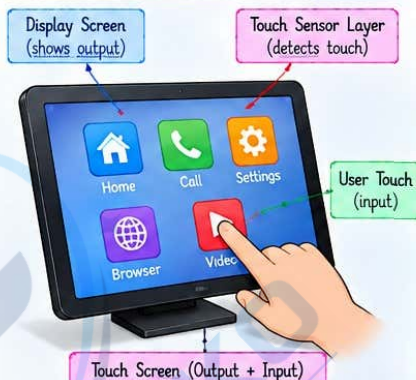
1) Introduction

- A touch screen is an output device that displays visual information and at the same time acts as an input device by detecting touch.
- It allows the user to interact with the computer by touching icons, buttons or menus directly on the screen.
- It is widely used in smartphones, tablets, ATMs, kiosks, laptops and interactive display systems.
- Touch screens make the system user-friendly, fast and interactive.
- They eliminate the need for a separate keyboard or mouse in many applications.

Example :

Using a touch screen to open an app, draw, or type on a virtual keyboard.

2) Diagram of a Touch Screen



3) How a Touch Screen Works?

- The screen displays images, icons and information (output).
- When the user touches the screen, the touch sensor detects the touch input (change in electrical signal).
- The position of the touch is calculated by a controller.
- The controller sends this input to the computer.
- The operating system processes the input and performs the corresponding action (e.g., open app, select option).
- The result is displayed on the screen again (output).

Thus, a touch screen works as both an output and an input device.

4) Types of Touch Screens

- Resistive Touch Screen**
 - Made of two flexible layers.
 - Works by pressure.
 - Used in ATMs, POS systems.
- Capacitive Touch Screen**
 - Uses electrical properties of human body.
 - More sensitive and accurate.
 - Used in smartphones, tablets.
- Infrared Touch Screen**
 - Uses infrared light beams.
 - Detects touch when a beam is interrupted.
 - Used in large displays, kiosks.
- Surface Acoustic Wave (SAW)**
 - Uses ultrasonic waves.
 - High clarity and durability.
 - Used in public information systems.



5) Components of a Touch Screen

Component	Function
Display Panel	Shows the output (text, images, videos).
Touch Sensor Layer	Detects the touch input.
Controller	Processes the touch signals.
Interface	Connects to the computer (e.g., USB, HDMI, etc.).
Protective Layer	Provides durability and prevents damage.

6) Applications of Touch Screen

- Smartphones and tablets.
- Laptops and 2-in-1 devices.
- ATMs and banking terminals.
- Self-service kiosks (railway, airport, malls).
- Interactive whiteboards in education.
- Point of Sale (POS) systems in shops.
- Medical devices in hospitals.
- Industrial and control systems.
- Gaming and entertainment devices.

7) Advantages

- ✓ Easy to use and user-friendly.
- ✓ Saves space (no separate keyboard or mouse).
- ✓ Provides fast and direct interaction.
- ✓ More interactive and engaging.
- ✓ Useful in public places and self-service systems.
- ✓ Supports multitouch (multiple fingers).
- ✓ Widely used in modern devices.
- ✓ Improves productivity and accessibility.

8) Disadvantages

- ✗ More expensive than traditional input/output devices.
- ✗ Can get scratches and fingerprints.
- ✗ May not work properly with wet hands (in some types).
- ✗ Prolonged use can cause finger fatigue.
- ✗ Not as precise as a mouse for detailed work.
- ✗ Can be damaged if the screen is cracked.
- ✗ Requires regular cleaning.
- ✗ Performance may vary in extreme temperatures.

9) Comparison: Touch Screen vs Traditional Devices

Feature	Touch Screen	Monitor + Mouse/Keyboard
Function	Output + Input	Separate output and input
Space	Saves space	Requires more space
Interaction	Direct touch	Indirect (via mouse/keyboard)
Portability	More portable	Less portable
Usage	Common in modern devices	Traditional and widely used
Cost	Expensive	More affordable
User Experience	More interactive	Less interactive

10) Key Points to Remember

- A touch screen is both an output and an input device.
- It displays information and detects touch.
- Common types: Resistive, Capacitive, Infrared, SAW.
- It is widely used in smartphones, ATMs, kiosks, etc.
- Capacitive touch screens are the most common.
- It makes computer systems more user-friendly and interactive.
- It requires proper care and maintenance.
- Touch screens are an important part of modern technology.

"Touch the Screen
Touch the Future!"



Interact • Explore • Create
All at Your Fingertips!"

Monitors: VDU, LCD & LED

"Better Display
Brighter Ideas
Clearer Tomorrow!"

A monitor is a visual output device that displays text, images, videos and other information generated by the computer.

1) Introduction

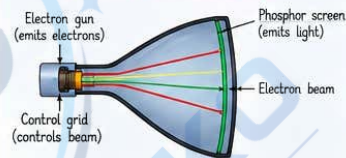
- A monitor is an output device that displays processed data from the computer in the form of text, images, graphics and video.
- It allows the user to see and interact with the information.
- Monitors are also called Visual Display Units (VDU).
- There are different types of monitors based on the display technology, such as CRT (VDU), LCD and LED.
- Monitors are an essential part of modern computing systems and are widely used in homes, offices, schools, industries and entertainment.

Example : Displaying a document, watching a movie, playing a game, or browsing the internet.

2) What is a VDU ?

- VDU stands for Visual Display Unit.
- It is the traditional type of monitor that uses Cathode Ray Tube (CRT) technology.
- It displays images by using electron beams that strike a phosphor-coated screen.
- VDUs were widely used in earlier computers but are now less common due to their large size and higher power consumption.

Working of CRT (VDU)

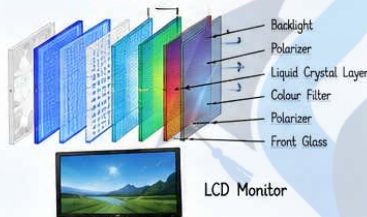


When electron beams hit the phosphor screen, small dots glow to form images.

3) What is an LCD Monitor ?

- LCD stands for Liquid Crystal Display.
- It uses liquid crystals that change their alignment when an electric current is applied, allowing light to pass through and create images.
- LCDs are flat, lightweight and consume less power than CRT monitors.
- They provide good image quality and are widely used in laptops, desktops and TVs.

Structure of an LCD



Liquid crystals control the amount of light passing through each pixel to produce images.

4) What is an LED Monitor ?

- LED stands for Light Emitting Diode.
- An LED monitor is an improved version of LCD that uses LED backlighting instead of CCFL (cold cathode fluorescent lamps).
- LEDs provide brighter images, higher contrast and better energy efficiency.
- They are thinner, lighter and more environment-friendly than LCD and CRT monitors.
- LED monitors are widely used in modern computers, TVs, smartphones and large displays.

LED Backlight Technology



LEDs are used as a backlight to provide light for the LCD panel, resulting in brighter and clearer images.

5) Comparison Table

Feature	VDU (CRT)	LCD	LED
Full Form	Visual Display Unit (Cathode Ray Tube)	Liquid Crystal Display	Light Emitting Diode
Technology	Electron beam strikes phosphor screen	Uses liquid crystals with fluorescent backlight (CCFL)	Uses LED backlighting
Size & Weight	Bulky and heavy	Thin and light	Thinner and lighter
Power Consumption	High	Low	Very low
Image Quality	Good (but lower resolution)	Better (sharp and clear)	Best (high brightness and contrast)
Heat Generation	More heat	Less heat	Very less heat
Lifespan	Shorter	Longer	Longest
Cost	Low (earlier)	Moderate	Slightly higher
Common Uses	Older computers, TVs	Laptops, monitors, TVs	Modern monitors, TVs, gaming, large displays

6) Advantages and Disadvantages

Type	Advantages	Disadvantages
VDU (CRT)	- Cheap (earlier) - Good color reproduction - Wide viewing angles	- Bulky and heavy - High power consumption - More heat generation - Lower resolution - Not environment-friendly
LCD	- Thin and lightweight - Low power consumption - Clear and sharp images - Less heat generation - Affordable	- Limited viewing angles - Lower contrast than LED - Uses fluorescent backlight (CCFL)
LED	- Bright and high contrast images - Very low power consumption - Thin and lightweight - Long lifespan - Environment-friendly	- Costlier than LCD - May cause eye strain if brightness is too high

7) Applications

- Used in desktops, laptops and all-in-one computers.
- Widely used in offices, schools, colleges and industries.
- Essential for graphic designing, photo and video editing.
- Used in gaming and entertainment.
- Used in medical, educational and public information systems (kiosks, ATMs, digital signage).

8) Key Points to Remember

- A monitor is an output device.
- VDU uses CRT technology.
- LCD uses liquid crystals.
- LED uses LED backlighting.
- LED monitors are more efficient and provide better display quality than LCD and VDU.
- Monitors are essential for visual output and user interaction.
- Modern monitors are thinner, lighter, clearer and more energy-efficient.



"Same Information...
Clearer, Brighter,
Better with
Every Generation!"

Printers : Types of Printers

"Your Ideas Into Reality!"

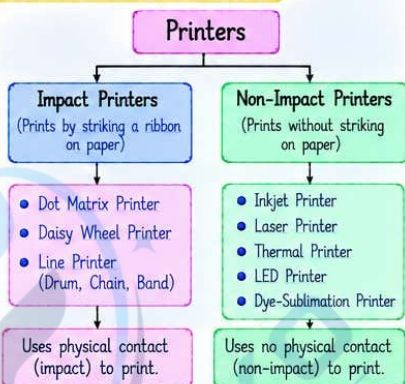
A printer is an output device that produces a permanent (hard) copy of text, images or graphics on paper or other media.

1) Introduction

- A printer is an output device that converts digital information from a computer into a printed (hard) copy on paper or other materials.
- It produces permanent output, which can be text, images, graphics or a combination of these.
- Printers are widely used in homes, offices, schools, banks, hospitals and industries.
- Printers make it easy to share and keep a physical record of important information.

Example : Printing a report, photograph, invoice, ID card, etc.

2) Classification of Printers



3) Impact Printers

a) Dot Matrix Printer

- Prints characters and images as a pattern of dots.
- Uses a print head with small pins that strike an ink ribbon.
- Can print on multi-part (carbon) paper.
- Low cost and durable, but lower print quality and noisy.
- Used in banks, billing systems, and where multiple copies are needed.



Dot Matrix Printer

b) Daisy Wheel Printer

- Uses a wheel with pre-formed characters (like a daisy).
- Good quality text output.
- Slow speed and noisy.
- Used for word processing in earlier days.



Daisy Wheel Printer

c) Line Printers (Drum, Chain, Band)

- Prints an entire line at a time.
- Very high speed and used for large volume printing.
- Common types: Drum Printer, Chain Printer, Band Printer.
- Mostly used in industries and large organisations.



Line Printer

4) Non-Impact Printers

a) Inkjet Printer

- Sprays tiny droplets of liquid ink onto paper.
- Produces high-quality color prints.
- Useful for home and office use.
- Lower speed and higher cost per page compared to laser printers.



Inkjet Printer

b) Laser Printer

- Uses laser technology and toner (powder ink).
- Produces high-speed, high-quality prints.
- Suitable for large volume printing.
- Common in offices and businesses.



Laser Printer

c) Thermal Printer

- Uses heat to transfer ink onto paper.
- Commonly used in POS machines, ATMs and receipt printers.
- Fast, silent and low maintenance.



Thermal Printer

d) LED Printer

- Works similar to a laser printer but uses LEDs instead of a laser beam.
- Faster, more efficient and reliable.
- Produces excellent print quality.



LED Printer

e) Dye-Sublimation Printer

- Uses heat to transfer dye onto paper.
- Produces high-quality, long-lasting color prints (mainly photos).
- Used in photo printing, ID cards, and professional printing.



Dye-Sublimation Printer

5) Comparison: Impact vs Non-Impact Printers

Feature	Impact Printers	Non-Impact Printers
Printing Method	Uses physical contact	Uses no physical contact
Print Quality	Lower	Higher
Speed	Generally slower (except line printers)	Faster (especially laser printers)
Noise	Noisy	Silent or less noisy
Cost	Lower	Higher
Ability to Print Multi-copy	Yes	Generally No
Examples	Dot Matrix, Daisy Wheel, Line Printer	Inkjet, Laser, Thermal, LED, Dye-Sublimation

6) Advantages and Disadvantages

Type	Advantages	Disadvantages
Dot Matrix	Low cost, durable, can print multi-copy	Low print quality, noisy, slow
Daisy Wheel	Good quality text	Slow, noisy, limited to text only
Line Printer	Very high speed, used for bulk printing	Expensive, noisy, large size
Inkjet	High quality color prints, cheap initial cost	Slower, ink is costly, not ideal for large volume
Laser	Fast, high quality, low cost per page	Higher initial cost
Thermal	Fast, silent, low maintenance	Limited to special paper, mainly used for receipts
LED	High speed, reliable, excellent quality	Higher cost
Dye-Sublimation	Best for photo quality, long-lasting prints	Expensive, slower, mainly for photo printing

7) Applications of Printers

- Printing documents, reports, letters, invoices.
- Used in offices, schools, colleges and banks.
- Printing photographs and color images.
- Used in hospitals for patient reports.
- Used in shops, malls and restaurants (receipts).
- Used in industries for labels and barcodes.
- Printing ID cards, tickets and certificates.
- Useful in home and personal use.

8) Key Points to Remember

- A printer is an output device.
- Two main types: Impact and Non-Impact.
- Impact printers use physical contact.
- Non-impact printers do not use physical contact.
- Common types: Dot Matrix, Daisy Wheel, Line, Inkjet, Laser, Thermal, LED, Dye-Sublimation.
- Printers are essential for producing hard copies.
- Choose a printer based on speed, quality, cost and purpose.

"From Digital to Physical... Printers Make It Real!"

"Print Today
A Better Tomorrow"



Impact Printers

"Old
Technology...
Still Reliable!"

Impact printers are a type of printer that create characters and images by physically striking an inked ribbon against paper.

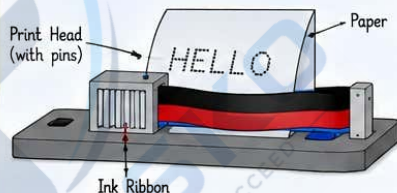
1) Introduction

- Impact printers are output devices that produce printed characters or images by striking a metal pin, hammer or type element against an inked ribbon, which in turn hits the paper.
- They work on the principle of mechanical impact.
- They are generally slower, noisier and produce lower quality output compared to non-impact printers, but are durable, reliable and cost-effective.
- Impact printers can print on multi-part forms (carbon copies), which is not possible with most non-impact printers.
- They are widely used in banks, railway reservations, billing, accounting and other industries where multiple copies are required.

Example : Printing bills, invoices, receipts, challans, etc.

2) Working Principle

- A print command is sent from the computer to the printer.
- The print head contains pins or a type element that moves forward and strikes an inked ribbon.
- The ribbon transfers ink onto the paper, forming characters or images.
- The paper is advanced and the next character is printed in the same way.
- This process is repeated until the complete document is printed.



Characters are formed by striking pins against an inked ribbon, which transfers ink to the paper.

3) Types of Impact Printers

(a) Dot Matrix Printer

- Most common type of impact printer.
- Uses a print head with small pins (9, 18 or 24 pins) that strike an inked ribbon to form characters as a pattern of dots.
- Can print text and simple graphics.
- Used in offices, banks, railway reservation systems, billing, etc.



Dot Matrix Printer

(b) Daisy Wheel Printers

- Uses a wheel with raised characters (like a daisy).
- Each petal of the wheel has a character.
- When the required character is selected, the wheel rotates and the character strikes the inked ribbon and paper.
- Produces high quality, letter-like print.
- Generally used for personal and small business use.



Daisy Wheel Printer



Daisy Wheel
(Petals with Characters)

(c) Line Printer

- Prints an entire line of text at a time.
- Very high speed and used for large volume printing.
- Common types: Drum Printer, Chain Printer, Band Printer.
- Used in industries, bulk printing and data processing centers.

(d) Character (Shuttle) Printer

- Uses a print head that moves horizontally (shuttle) across the page.
- Characters are printed one at a time.
- Slower compared to line printers.
- Used in early computers and terminals.



Character (Shuttle) Printer

4) Comparison of Impact Printers

Feature	Dot Matrix	Daisy Wheel	Line Printer	Character (Shuttle)
Printing Method	Pins strike ribbon (to form dots)	Character wheel strikes ribbon	Prints a whole line at a time	Print head moves horizontally
Print Quality	Moderate	High (letter-like)	Moderate	Low to moderate
Speed	Slow	Slow	Very high	Slow
Graphics Support	Yes (simple)	No	Limited	No
Noise Level	Noisy	Noisy	Very noisy	Noisy
Cost	Low	Moderate	High	Low
Typical Use	Offices, billing	Personal, small business	Industries, bulk printing	Early computers, terminals

5) Advantages

- Can print on multi-part forms (carbon copies).
- Robust and durable.
- Reliable and long-lasting.
- Works well in harsh environments.
- Lower cost per page.
- Widely used and well-proven technology.
- Requires less maintenance compared to some non-impact printers.

6) Disadvantages

- Lower print quality compared to non-impact printers (e.g., laser, inkjet).
- Noisy operation.
- Slower printing speed (except line printers).
- Limited color support.
- Not suitable for high-quality graphics and images.
- Wear and tear of mechanical parts.
- Bulky design (in some types).
- Higher power consumption (in line printers).

7) Applications

- Banks and financial institutions (cheques, passbooks).
- Railway and airline reservation systems.
- Billing and accounting systems.
- Printing receipts, invoices and challans.
- Industrial and warehouse environments.
- Government offices.
- Places where multiple copies are required.

"Impact Printers - When Copies Matter!"

"Reliable Prints
for Real Work!"

Non - Impact Printers

Non-impact printers are output devices that produce printed characters and images without using any physical contact with the paper.

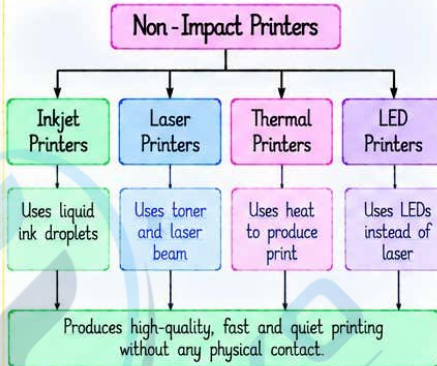
"No Strike...
Just Smart Printing!"

1) Introduction

- Non-impact printers are output devices that print without physically striking the paper.
- They use various technologies such as ink, toner, heat or light to produce high-quality text and images.
- They are faster, quieter and produce better print quality than impact printers.
- They are widely used in homes, offices, schools, hospitals, industries and commercial printing.
- Non-impact printers can print text, graphics, images and color documents.
- They are more expensive than impact printers but offer better features and performance.

Example : Printing reports, photos, color documents, brochures, etc.

2) Types of Non-Impact Printers



3) Inkjet Printer

- Uses tiny droplets of liquid ink to print on paper.
- Ink is sprayed through small nozzles onto the paper.
- Can print both text and high-quality color images.
- Suitable for home and small office use.
- Lower initial cost but higher cost per page.
- Commonly used for photos, color printing and everyday documents.



Inkjet Printer

4) Laser Printer

- Uses a laser beam and toner (powdered ink) to produce print.
- Laser beam creates an electrostatic image on a drum, toner sticks to it and then transfers to paper.
- Produces high-speed, high-quality and sharp prints.
- Suitable for offices and large volume printing. Lower cost per page but higher initial cost.
- Commonly used for documents, reports and professional printing.



Laser Printer

5) Thermal Printer

- Uses heat to transfer ink onto special thermal paper.
- No ink or toner is required.
- Produces fast and quiet printing.
- Commonly used in POS machines, ATMs, billing systems and receipt printers.
- Low maintenance but requires special thermal paper.



Thermal Printer

6) LED Printer

- Works similar to laser printer but uses LED (Light Emitting Diode) array instead of laser beam.
- LEDs create the electrostatic image on the drum.
- Provides high-speed, high-quality and reliable printing.
- More energy efficient and has fewer moving parts.
- Commonly used in offices and industries for large volume printing.



LED Printer

7) Comparison Table

Feature	Inkjet	Laser	Thermal	LED
Printing Technology	Liquid ink droplets	Laser beam + toner	Heat on thermal paper	LED array + toner
Print Quality	Good (color)	Excellent (sharp text)	Good (text/receipts)	Excellent (sharp & clear)
Printing Speed	Slow to moderate	Fast	Very fast	Fast
Color Printing	Yes	Yes	Limited (often no)	Yes
Cost per Page	High	Low	Low	Low
Initial Cost	Low	High	Moderate	High
Common Use	Home, photos, small office	Office, bulk documents	Receipts, POS, ATMs	Office, large printing

8) Advantages

- ✓ High-quality and clear printing.
- ✓ Faster printing speed.
- ✓ Quieter operation.
- ✓ Can print text, images and color documents.
- ✓ More reliable and durable.
- ✓ Suitable for high-volume printing (in laser and LED).
- ✓ Wide range of models and sizes available.
- ✓ Produces professional and attractive output.

9) Disadvantages

- ✗ Higher initial cost (except inkjet).
- ✗ Cost per page can be high (especially inkjet).
- ✗ Requires special materials (e.g. thermal paper).
- ✗ Not suitable for multi-part forms (carbon copies).
- ✗ Maintenance and cartridge replacement can be expensive.
- ✗ Some models may be costly for small businesses.
- ✗ Ink can dry out if not used for a long time (inkjet).

10) Applications

- Homes and personal use (photos, documents).
- Offices and schools (reports, assignments).
- Banks and financial institutions (receipts, passbooks).
- Retail stores and supermarkets (billing receipts).
- Hospitals and medical records.
- Industries and warehouses (labels, barcodes).
- Government offices and public services.
- Professional printing and graphic design.

"Print Without Limits
Create a Brighter Tomorrow!"

Non - Impact Printers → Better Quality → A Smarter Future!



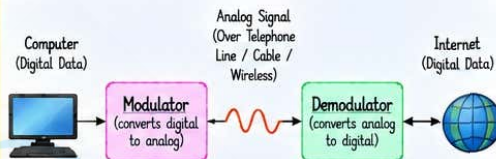
A modem is a device that modulates and demodulates signals to enable communication between a computer and the Internet over telephone lines or other communication channels.

1) Introduction

- A **modem (Modulator-Demodulator)** is a device that allows a computer to send and receive digital data over analog communication lines such as telephone lines, cable or wireless networks.
- It converts digital signals from the computer into analog signals for transmission (modulation) and converts received analog signals back into digital form (demodulation).
- It enables internet connectivity and communication between distant computers.
- Modems are widely used in homes, offices, businesses and ISPs (Internet Service Providers).

Example : Connecting to the Internet, sending emails, online browsing, etc.

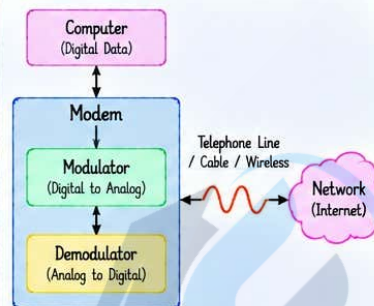
2) Working of a Modem



Working Steps:

- Computer sends digital data to the modem.
- Modulator converts digital data into an analog signal.
- The analog signal is transmitted over a communication channel (telephone line, cable, fiber or wireless).
- At the receiving end, the demodulator converts the analog signal back into digital data.
- The data is then sent to the destination computer or the Internet.

3) Block Diagram of a Modem

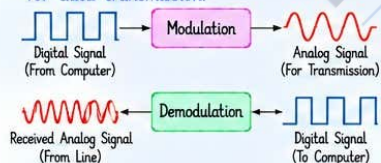


4) Types of Modems

- Dial-up Modem**
Uses telephone lines and dials a number to connect to the internet. Speed is low (up to 56 Kbps).
- DSL Modem**
Uses existing telephone lines (DSL - Digital Subscriber Line). Faster than dial-up.
- Cable Modem**
Uses cable TV network for internet access. Provides high-speed internet.
- Wireless Modem**
Uses mobile networks (3G/4G/5G) or Wi-Fi for internet connectivity.
- Fiber Optic Modem (ONT)**
Used in fiber optic networks. Provides very high speed and reliability.
- Satellite Modem**
Used for satellite internet communication (e.g., in remote areas).

5) Modulation and Demodulation

- Modulation:** The process of converting digital data into an analog signal suitable for transmission over a communication channel.
- Demodulation:** The process of converting the received analog signal back into digital data.
- Modulation techniques used: ASK (Amplitude Shift Keying), FSK (Frequency Shift Keying), PSK (Phase Shift Keying) for data transmission.



6) Types of Connections

- Wired Modem**
Uses telephone line, cable or fiber optic cables.
- Wireless Modem**
Uses radio waves (Wi-Fi, mobile networks).
- External Modem**
Separate device connected via USB or serial port.
- Internal Modem**
Installed inside the computer (e.g., PCI card). (less common now)
- USB Modem**
Portable modem connected through a USB port (e.g., dongles).
- Router with Modem**
Many routers come with a built-in modem (modem + router).

7) Advantages and Disadvantages

Advantages

- ✓ Enables internet connectivity.
- ✓ Allows communication over long distances.
- ✓ Supports various types of connections (dial-up, DSL, cable, wireless, etc.).
- ✓ Easy to install and use.
- ✓ Widely available and cost-effective.
- ✓ Modern modems provide high-speed internet.

Disadvantages

- ✗ Dial-up modems are slow.
- ✗ Dependent on communication infrastructure (telephone lines, cable, etc.).
- ✗ Can be affected by noise and signal interference.
- ✗ Security risks if not properly configured.
- ✗ Performance may vary based on network type and service provider.

8) Applications of Modem

- Internet access (browsing, streaming, downloading).
- Email communication.
- Online education and work-from-home.
- Video conferencing.
- E-commerce and online banking.
- Connecting remote locations.
- Used in homes, offices, schools, hospitals, industries, and remote areas.

9) Key Points to Remember

- Modem stands for Modulator-Demodulator.
- It converts digital ↔ analog signals.
- Essential for internet connectivity.
- Types: Dial-up, DSL, Cable, Wireless, Fiber, Satellite.
- Modern routers often include a modem.
- Speed and performance depend on the type of modem and network.
- Used globally for communication and data transfer.

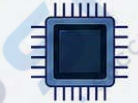
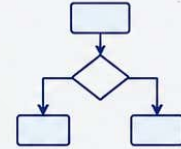
"Modem - Turning Signals into Possibilities!"



"Stay Connected Stay Ahead!"

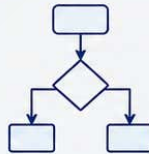


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Unit - IV

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Unit - II

I/O Devices

Operating System

"Manage Resources...
Make Computing Easier!"

An Operating System (OS) is system software that acts as an interface between the user and the computer hardware.

1) Introduction

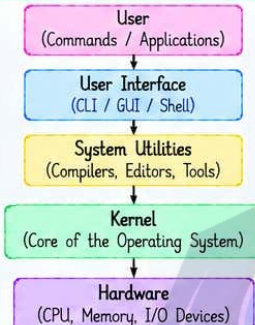
- An Operating System is a software that manages computer hardware and software resources.
- It provides a platform for users to run applications and acts as an interface between the user and the hardware.
- It controls the execution of programs and manages input/output operations, memory, files, and devices.
- Examples of Operating Systems: Windows, Linux, macOS, Android, iOS.

Example: When you start a computer, the OS loads into the memory and provides a user interface (e.g., desktop) to work with.

2) Functions of an Operating System

- Process management (creating, scheduling, terminating processes)
- Memory management (allocating and deallocating memory)
- File management (creating, deleting, organizing files)
- Device management (controlling I/O devices)
- Security and user management (authentication and access control)
- Providing user interface (CLI or GUI)
- Error detection and handling
- Resource allocation and accounting
- Networking and communication support

3) Components of an Operating System



4) Types of Operating Systems

1. **Batch Operating System**
Executes jobs in batches without user interaction.
2. **Time-Sharing Operating System**
Allows multiple users to use the system simultaneously (e.g., UNIX).
3. **Real-Time Operating System**
Provides immediate response to inputs (e.g., in industrial and medical devices).
4. **Distributed Operating System**
Manages a group of interconnected computers as a single system.
5. **Network Operating System**
Provides network services like file sharing, printing, communication (e.g., Windows Server).
6. **Mobile Operating System**
Designed for mobile devices (e.g., Android, iOS).
7. **Embedded Operating System**
Used in embedded systems (e.g., smart TVs, appliances).

5) User Interface in Operating System

Command Line Interface (CLI)

- Text-based interface.
- Users type commands to perform tasks.
- Examples: MS-DOS, Linux Shell.

```
C:\>dir _
```

Graphical User Interface (GUI)

- Uses icons, menus, and mouse for interaction.
- Easy to use and user-friendly.
- Examples: Windows, macOS, Linux (GNOME/KDE).



6) Process Management

- A process is a program in execution.
- The OS is responsible for creating, terminating and scheduling processes.
- It maintains a Process Control Block (PCB) for each process.
- Handles process states: New, Ready, Running, Waiting, Terminated.

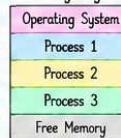
Process States



7) Memory Management

- Manages the computer's main memory (RAM).
- Allocates and deallocates memory to processes.
- Keeps track of free and used memory.
- Uses techniques like paging and segmentation.

Memory Layout



The OS ensures efficient and fair use of memory among multiple processes.

7) File Management

- Handles creation, deletion, reading, writing and organizing files.
- Maintains file directories and metadata.
- Provides file system (e.g., NTFS, FAT, ext4).
- Controls file access permissions.

9) Device Management

- Controls and manages I/O devices such as keyboard, mouse, monitor, printer, etc.
- Uses device drivers to communicate with hardware.
- Manages device allocation and scheduling.
- Handles input/output operations efficiently.

I/O Devices and OS



10) Advantages and Disadvantages

Advantages

- ✓ Provides easy and user-friendly interface.
- ✓ Efficiently manages hardware and software resources.
- ✓ Supports multitasking and multiuser environment.
- ✓ Improves system security.
- ✓ Hides hardware complexity from the user.
- ✓ Provides file and device management.
- ✓ Supports networking and communication.

Disadvantages

- ✗ Complex to design and maintain.
- ✗ Consumes system resources (memory and CPU).
- ✗ Can be expensive.
- ✗ If the OS fails, the entire system can be affected.
- ✗ Requires regular updates and patches.
- ✗ May have security vulnerabilities.
- ✗ Some advanced features may be difficult for beginners.

Real Life Examples

- Windows - personal computers and offices.
- Linux - servers, education and open source.
- Android - smartphones and tablets.
- macOS - Apple devices.
- UNIX - multiuser environments.
- RTOS - used in ATMs, traffic control, medical devices.
- Embedded OS - used in smart TVs, washing machines, IoT devices.

★ Key Points to Remember

- ★ Operating System is an interface between user and hardware.
- ★ Manages processes, memory, files and devices.
- ★ Types: Batch, Time-sharing, Real-time, Distributed, Network, Mobile, Embedded.
- ★ Provides CLI and GUI.
- ★ Kernel is the core component.
- ★ Examples: Windows, Linux, macOS, Android, iOS.
- ★ Main goal: Efficient, secure and easy use of computer resources.

"Operating System - The Brain Behind Every Computer!"

"Better Systems...
Brighter Possibilities!"

Unit - II

I/O Devices

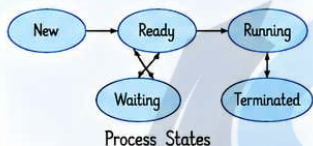
Functions of Operating System

"A Good OS
Keeps Everything
Running Smoothly!"

An Operating System performs various functions to manage computer resources and provide a convenient environment for users to run programs.

1) Process Management

- Creates, schedules and terminates processes.
- Allocates CPU to processes using scheduling algorithms.
- Maintains Process Control Block (PCB)** for each process.
- Handles process states (New, Ready, Running, Waiting, Terminated).
- Ensures smooth multitasking and improves CPU utilization.



2) Memory Management

- Keeps track of the computer's main memory (RAM).
- Allocates and deallocates memory to processes.
- Ensures efficient and fair use of memory.
- Uses techniques like paging, segmentation and virtual memory.
- Protects memory from unauthorized access.



Main Memory (RAM)

3) File Management

- Handles creation, deletion, reading, writing and organizing files.
- Maintains file directories and metadata (file name, size, type, date, permissions).
- Provides file system (e.g., NTFS, FAT, ext4).
- Controls file access permissions.
- Ensures data integrity and backup support.



Files and Directories

4) Device Management

- Controls and manages all I/O devices such as keyboard, mouse, monitor, printer, disk, etc.
- Uses device drivers to communicate with hardware.
- Manages device allocation and scheduling.
- Handles input/output operations efficiently.
- Provides buffering, spooling and interrupt handling.



I/O Devices

5) Security and Protection

- Protects system resources from unauthorized access.
- Manages user authentication (username and password).
- Provides access control (permissions to files, devices, etc.).
- Ensures data confidentiality, integrity and availability.
- Detects and prevents malicious activities (viruses, malware).
- Supports user roles (admin, guest, normal user).



Keeps
your data
safe!

6) User Interface Management

- Provides an interface for users to interact with the system.
- Can be Command Line Interface (CLI) or Graphical User Interface (GUI).
- Makes the system easy to use and user-friendly.
- Interprets user commands and executes them.
- Examples: Windows (GUI), Linux (shell), macOS (GUI).



CLI User Interface GUI

7) Resource Allocation

- Allocates resources like CPU, memory, devices and files to different processes and users.
- Ensures fair and efficient use of resources.
- Manages conflicts when multiple processes request the same resource.
- Uses scheduling and queueing techniques.



8) Error Detection and Handling

- Detects errors in hardware and software.
- Handles errors gracefully without crashing the system.
- Provides error messages to users.
- Takes corrective actions and maintains system stability.
- Examples: disk read/write error, memory error, device failure.



Keeps the system stable!

9) Networking and Communication

- Provides network services for communication between computers.
- Manages data sharing and resource sharing over networks.
- Supports protocols like TCP/IP.
- Helps in client-server communication.
- Used in distributed systems and internet services.



10) Accounting and Performance Monitoring

- Keeps track of resource usage by users and processes.
- Maintains logs and statistics (CPU usage, memory usage, disk usage, etc.).
- Helps in performance tuning and optimization.
- Used for billing in multi-user environments (e.g., organizations, cloud services).



Monitor
Manage
Improve
Performance!

11) Provides Execution Environment

- Loads and runs programs in memory.
- Provides necessary services and system calls.
- Acts as a bridge between user programs and hardware.
- Ensures programs run smoothly and efficiently.

"An Operating System doesn't just run programs,
It makes everything work together!"

Key Takeaway

- The OS manages resources, provides security, ensures smooth execution, and makes the computer easy to use.
- Without an OS, a computer is just a collection of hardware resources.
- Examples: Windows, Linux, macOS, Android, iOS.

Batch Mode is a type of operating system in which similar jobs are collected together, grouped into batches and executed automatically without user interaction.

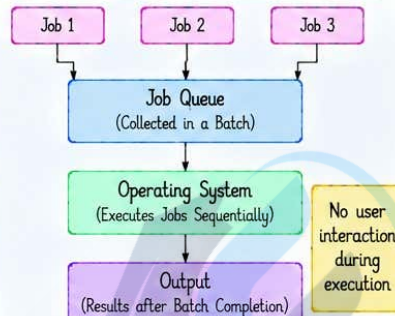
1) Introduction

- Batch Mode is one of the earliest and simplest types of operating systems.
- In this mode, jobs with similar requirements are collected and grouped into a batch.
- The batch is then processed sequentially without any user interaction.
- Users submit their jobs (programs and data) to the system, and the OS executes them one by one.
- This mode was widely used in mainframe and early computers.

2) How Batch Mode Works ?

- 1 Users submit jobs (programs and data) to the system.
- 2 Jobs are stored in a job queue.
- 3 The operating system groups similar jobs into a batch.
- 4 The batch is executed sequentially without user interaction.
- 5 When one job finishes, the next job in the batch is executed.
- 6 Output is collected and provided to the user after all jobs in the batch are completed.

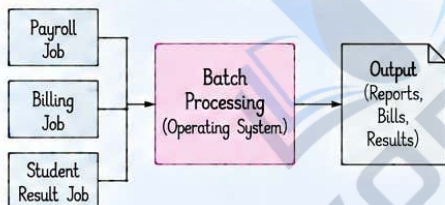
3) Block Diagram of Batch Processing



4) Characteristics

- Similar jobs are grouped together.
- Jobs are executed sequentially.
- No user interaction during execution.
- Uses job queue and spooling.
- Efficient for handling large number of similar jobs.
- Increases CPU utilization.
- Reduces idle time.
- Provides output after batch completion.

5) Example of Batch Processing



All jobs are collected and executed as a batch without user interaction.

6) Advantages

- ✓ Efficient use of CPU and resources.
- ✓ Reduces idle time.
- ✓ Suitable for large and repetitive tasks.
- ✓ High throughput.
- ✓ Easy to manage similar jobs.
- ✓ Useful for organizations (banks, railways, payroll, etc.).

7) Disadvantages

- ✗ No user interaction during execution.
- ✗ Results are available only after all jobs are completed.
- ✗ Difficult to debug errors.
- ✗ Not suitable for time-critical or interactive tasks.
- ✗ If one job fails, the entire batch may be affected.
- ✗ Long waiting time for output.

8) Applications

- Payroll processing.
- Bank statement generation.
- Billing and invoice generation.
- Student result processing.
- Scientific data analysis.
- Government and organizational record processing.
- Used in mainframe and large computing environments.

9) Real-Life Example

- In a bank, thousands of customer transactions are collected and processed at night in batch mode.
- In a university, student results of all departments are processed together as a batch.
- In companies, monthly salary (payroll) is calculated in batch mode.

"Batch Mode - Many Jobs Together,
One Step Towards Efficiency!"

10) Key Points to Remember

- Batch mode groups similar jobs.
- Executes jobs sequentially.
- No user interaction.
- Output is given after batch completion.
- Used in early operating systems and mainframes.
- Best for large, repetitive, and non-interactive tasks.

Multiprogramming is an operating system technique in which more than one program is loaded into the main memory at the same time and the CPU switches between them to keep itself busy and improve overall system utilization.

1) Introduction

- Multiprogramming allows multiple programs to reside in memory simultaneously.
- It increases CPU utilization by switching between programs when one is waiting for I/O.
- It is the basis for modern operating systems and is used in batch systems and time-sharing systems.
- It reduces idle time of CPU and improves system throughput.

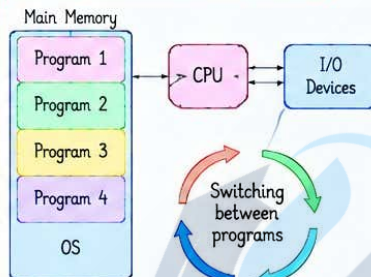
Example : In a computer, while one program is waiting for input, another program can use the CPU.

2) How Multiprogramming Works ?

- Multiple programs are loaded into main memory.
- The OS keeps track of their status using a process control block (PCB).
- When a program is waiting for I/O, the OS switches to another ready program.
- The CPU is kept busy with a different program instead of waiting.
- This cycle continues, increasing system utilization and throughput.

Key Idea : While one program is waiting for I/O, CPU can execute another program.

3) Diagram of Multiprogramming



4) Types of Multiprogramming

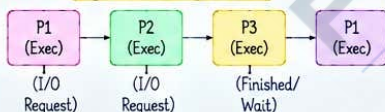
- Fixed Multiprogramming (Static)**
 - The number of programs in memory is fixed.
 - Uses a fixed partitioning of memory.
 - Simple but not flexible.
 - Common in early operating systems.
- Dynamic Multiprogramming**
 - The number of programs can be changed dynamically.
 - Uses variable partitioning (like virtual memory).
 - More efficient and flexible.
 - Used in modern operating systems.

5) Example Scenario

Consider a system with 3 programs (P1, P2, P3) loaded into memory.

- P1 is executing and then requests I/O.
- CPU switches to P2.
- P2 is executing and then requests I/O.
- CPU switches to P3.
- When P3 finishes or waits, CPU goes back to P1.
- This way, all programs make progress without wasting CPU time.

CPU Switching Example



6) Advantages

- Increases CPU utilization.
- Improves system throughput.
- Reduces idle time of CPU.
- Better use of system resources.
- Allows for multitasking environment.
- More efficient use of memory.
- Improves response time (in time-sharing systems).

7) Disadvantages

- More complex memory management.
- Need for OS overhead (context switching).
- Possible resource conflicts.
- Difficult to debug and maintain.
- Security and protection issues.

8) Comparison: Single vs Multiprogramming

Feature	Single Program	Multiprogramming
Programs in memory	One at a time	Multiple at a time
CPU utilization	Low	High
Idle time	More	Less
Throughput	Low	High
Complexity	Simple	Complex
Examples	Early OS	Modern OS (Batch, Time-sharing)

9) Real-Life Applications

- Batch operating systems (e.g., IBM mainframes).
- Servers handling multiple user requests.
- Cloud computing environments.
- Multitasking systems (Windows, Linux, macOS).

10) Key Takeaways

- Multiprogramming = multiple programs in memory.
- Keeps CPU busy and improves utilization.
- Types: Fixed and Dynamic.
- Used in batch and time-sharing systems.
- More complex but more efficient than single programming.



"Multiprogramming - Better Utilization, Higher Throughput!"

Unit - II

I/O Devices

Types : Time-Sharing

"Many Users
One System
Equal Opportunity!"

Time-Sharing is a type of operating system in which multiple users can interact with the computer system at the same time, and each user gets a small time slot (time quantum) of the CPU.

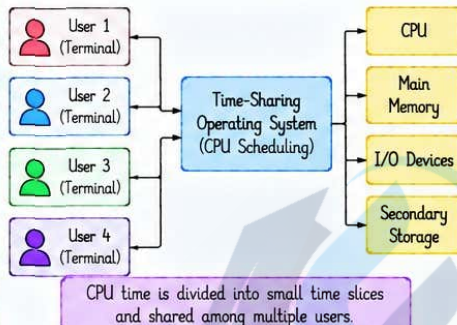
1) Introduction

- Time-Sharing is an extension of multiprogramming.
- It allows multiple users to use the same computer system simultaneously.
- The CPU time is divided into small time slices (time quantum) and each user gets a turn.
- The system switches quickly between users, giving the illusion that all users are executing their programs at the same time.
- It is mainly used in multi-user environments.

2) How Time-Sharing Works ?

- Multiple users log in to the system through terminals.
- Each user submits commands or programs.
- The CPU allocates a small time slice (time quantum) to each user.
- After the time slice expires, the CPU switches to another user (context switching).
- This process continues rapidly, giving interactive response to all users.
- The operating system manages memory, CPU scheduling and I/O for multiple users.

3) Diagram of Time-Sharing System

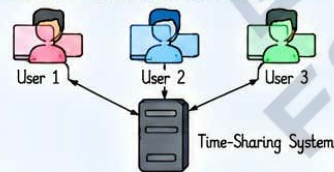


4) Characteristics

- Multiple users can use the system simultaneously.
- Each user gets a small time slice of CPU.
- Provides quick response time.
- Uses context switching.
- Requires memory management and CPU scheduling.
- Interactive and user-friendly.
- Increases CPU utilization.
- Supports multi-user environment.
- More complex than batch and multiprogramming.
- Requires advanced hardware and software support.

5) Example of Time-Sharing System

- UNIX and Linux operating systems.
- Mainframe computers.
- Multi-user systems in universities, research centers and organizations.
- Modern operating systems like Windows Server (multi-user mode).



6) Advantages

- ✓ Allows multiple users to work simultaneously.
- ✓ Provides quick and interactive response.
- ✓ Better CPU utilization.
- ✓ Reduces idle time.
- ✓ Efficient use of system resources (CPU, memory, I/O).
- ✓ Supports fairness (each user gets equal time).
- ✓ Suitable for large organizations and educational institutions.

7) Disadvantages

- ✗ More complex to design and implement.
- ✗ Requires advanced CPU scheduling and memory management.
- ✗ Overhead due to context switching.
- ✗ Requires powerful hardware and operating system software.
- ✗ Security and data protection become more critical.
- ✗ Performance may decrease if many users are active.
- ✗ Cost is higher compared to simpler systems.

8) Comparison with Other Types

Feature	Batch Processing	Multiprogramming	Time-Sharing
User Interaction	No	No	Yes
Users	One (at a time)	Multiple (no interaction)	Multiple (simultaneous)
Response Time	Slow	Moderate	Quick
CPU Utilization	Low	High	High
Main Goal	Process jobs in batches	Keep CPU busy	Provide interactive use to many users
Complexity	Simple	Moderate	High

9) Real-Life Examples

- College computer labs where many students log in at the same time.
- Office networks with shared servers.
- Cloud computing platforms (e.g., virtual machines).
- Online services like email, web servers, etc.

"Time-Sharing Brings People Together
on a Single System!"

10) Key Points to Remember

- Time-Sharing is an extension of multiprogramming.
- Allows multiple users to work simultaneously.
- CPU time is divided into small time slices.
- Uses context switching.
- Provides quick response time.
- Examples: UNIX, Linux, mainframe systems.
- Goal: efficient, interactive and fair use of system resources.

"Share
Time
Share Progress!"

Unit - II

I/O Devices

Types : Online Real Time

"Real Time
Real Response
Real Impact!"

Online Real Time is a type of operating system that processes data and responds to user requests immediately, as they occur, within a specified time limit.

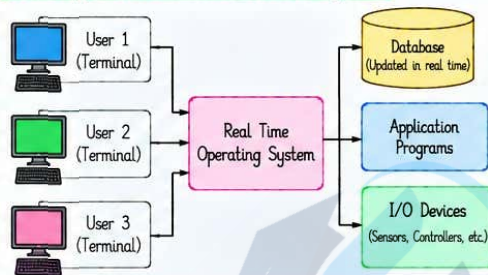
1) Introduction

- Online Real Time systems process data and respond to user requests immediately.
- Users interact with the system directly through terminals or networked devices.
- It provides instant access to data and up-to-date information at the time of request.
- Unlike batch or time-sharing systems, the main focus is immediate processing and quick response.
- Widely used in business, industrial, and critical applications where delay is not acceptable.

2) How Online Real Time Works ?

- 1 User sends a request through a terminal or online device.
- 2 The request is immediately passed to the CPU.
- 3 The operating system processes the request without delay.
- 4 Data is fetched/updated from the database or files in real time.
- 5 The result is instantly returned to the user.
- 6 The system continuously monitors inputs and responds within a specified time limit.

3) Diagram of Online Real Time System

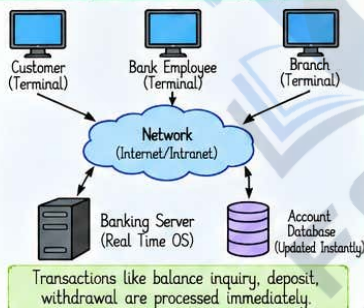


Processes user requests instantly and provides immediate response.

4) Characteristics

- Immediate response time.
- Direct user interaction (online).
- Processes transactions as they occur.
- Keeps data up-to-date at all times.
- High reliability and accuracy.
- Handles multiple users simultaneously.
- Requires powerful hardware and software.
- Used in time-critical and mission-critical environments.
- Examples: Banking, Airline reservation, Industrial control, etc.

5) Example: Online Banking System



6) Advantages

- ✓ Provides instant response.
- ✓ Keeps data accurate and up-to-date.
- ✓ Improves user satisfaction.
- ✓ Increases operational efficiency.
- ✓ Useful for critical and time-sensitive applications.
- ✓ Supports multiple users simultaneously.
- ✓ Helps in quick decision making.
- ✓ Reduces waiting time.

7) Disadvantages

- ✗ Requires high-performance and expensive hardware.
- ✗ Complex to design and maintain.
- ✗ Needs reliable and fast communication networks.
- ✗ System failure can cause serious problems.
- ✗ More security risks (online access).
- ✗ Requires skilled personnel.
- ✗ Not suitable for simple or small-scale applications.
- ✗ Continuous operation requires high maintenance.

8) Comparison with Other Types

Feature	Batch Mode	Time-Sharing	Online Real Time
User Interaction	No	Yes (multi-user)	Yes (immediate)
Response Time	Slow	Quick	Instant
Data Update	After batch completion	Periodic	Immediate (real time)
Purpose	Large, repetitive tasks	Multiple user access	Time-critical applications
Examples	Payroll, bills	Educational systems	Banking, reservations
Complexity	Simple	Moderate	High
Cost	Low	Moderate	High

9) Real-Life Applications

- Online banking systems (ATM, internet banking).
- Airline and railway reservation systems.
- Stock trading systems.
- Industrial process and machine control.
- Hospital and healthcare monitoring systems.
- Traffic control and signal systems.
- E-commerce websites (real-time order processing).

"In a Real Time System,
Every Second Counts!"

10) Key Points to Remember

- Online Real Time systems provide immediate response.
- Users interact directly with the system.
- Data is updated instantly.
- Used in time-critical and mission-critical environments.
- Examples: Banking, reservations, industrial control.
- More complex and costly than batch and time-sharing systems.
- Reliability, speed and security are very important.

"Faster
Smarter
Always Online!"

Unit - II

I/O Devices

Types : Distributed O.S.

A Distributed Operating System (Distributed O.S.) is a type of operating system that manages a group of independent computers, presents them as a single coherent system, and allows users to access resources and services from multiple machines transparently over a network.

"Many Computers
One System
Greater Possibilities!"

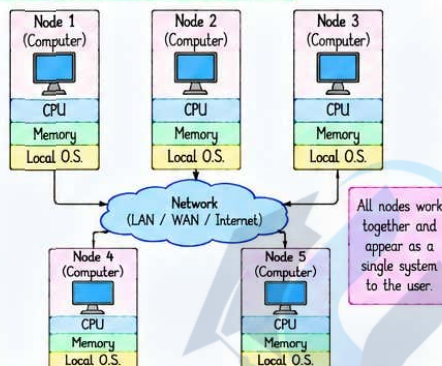
1) Introduction

- A Distributed Operating System (DOS) is designed to manage a collection of independent computers that work together and appear to the user as a single system.
- It hides the details of multiple machines and provides a unified environment.
- Resources such as files, processors, and devices can be shared across the network.
- It improves reliability, performance and scalability.
- Widely used in modern computing environments like cloud computing, clusters, and large organizations.

2) How Distributed O.S. Works ?

- 1 Multiple computers (nodes) are connected through a network.
- 2 Each node has its own CPU, memory and local O.S.
- 3 A Distributed O.S. manages the communication and coordination between nodes.
- 4 It provides transparent access to remote resources.
- 5 It uses middleware and communication protocols (e.g., RPC, message passing) for interaction.
- 6 The user sees the whole system as a single computer.

3) Diagram of Distributed O.S.



4) Characteristics

- Multiple independent computers working together.
- Transparency (location, access, migration, replication).
- Resource sharing across the network.
- High reliability and fault tolerance.
- Scalable and flexible.
- Better performance through load sharing.
- Heterogeneity (different hardware and OS can work together).
- Uses communication protocols for coordination.
- Appears as a single coherent system to the user.

5) Example of Distributed O.S.



Users at different locations can access and share resources as if it is a single system.

6) Advantages

- ✓ Resource sharing (files, devices, processing power).
- ✓ High reliability (failure of one node does not stop the system).
- ✓ Scalability (easy to add new nodes).
- ✓ Better performance (load sharing).
- ✓ Transparent access to resources.
- ✓ Supports heterogeneous systems.
- ✓ Cost effective (uses existing machines).
- ✓ Suitable for large organizations and global applications.

7) Disadvantages

- ✗ Complex design and implementation.
- ✗ Requires high-speed and reliable network.
- ✗ Security issues (data can be accessed from multiple locations).
- ✗ Difficult to maintain and manage.
- ✗ Failure in network can affect the entire system.
- ✗ Higher initial cost.
- ✗ Requires specialized hardware and software.
- ✗ Debugging and error detection is difficult.

8) Applications

- Distributed databases.
- Cloud computing (e.g., Google Cloud, AWS).
- File sharing systems (e.g., NFS).
- Distributed web services.
- Enterprise applications.
- Scientific and research computing.
- E-commerce and online services.
- Grid computing and cluster systems.
- Collaborative tools (e.g., Google Docs).

9) Real-Life Examples

- Google distributed file system (GFS).
- Amazon Web Services (AWS).
- Microsoft Azure.
- Distributed database systems (e.g., MongoDB, Cassandra).
- Online multiplayer games.
- Large corporate networks.

"Distributed Today,
A Stronger Tomorrow!"

10) Key Points to Remember

- Distributed O.S. manages a collection of computers as a single system.
- Provides transparency, reliability and scalability.
- Resources are shared over a network.
- More complex than centralized O.S.
- Widely used in modern applications like cloud computing and online services.
- Main goal: better resource utilization, reliability and performance.

"Share
Connect
Compute
Together!"

Unit - II

I/O Devices

Booting Process

"From Power On to a Ready System!"

Booting is the process of starting or restarting a computer and loading the operating system into the main memory (RAM) from a storage device.

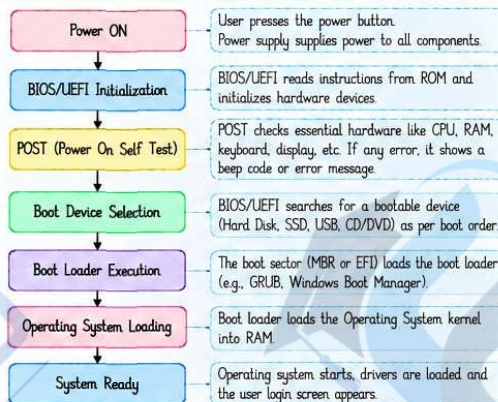
1) Introduction

- Booting is the initial process a computer follows when it is turned on.
- It loads the Operating System (OS) into the main memory so that the user can interact with the computer.
- The instructions for booting are stored in a small program called BIOS/UEFI in ROM.
- Booting can be a Cold Boot (when the system is powered on) or a Warm Boot (when the system is restarted).

2) Types of Booting

- Cold Booting**
 - System is turned on after complete power off.
 - Full hardware initialization is performed.
- Warm Booting**
 - System is restarted without turning off the power.
 - Faster than cold boot.
- Soft Booting**
 - Restarted using operating system options (e.g., Restart command).

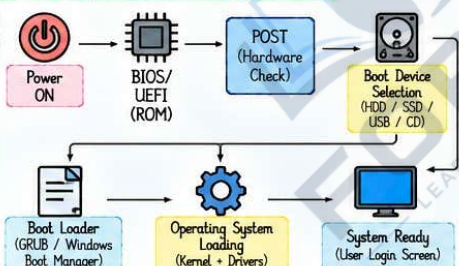
3) Step by Step Booting Process



4) Detailed Explanation

- Power ON**
 - User presses the power button.
 - SMPS supplies power to motherboard and other components.
- BIOS/UEFI Initialization**
 - BIOS/UEFI starts executing from ROM.
 - Initializes CPU, memory and basic I/O devices.
- POST (Power On Self Test)**
 - Tests RAM, CPU, keyboard, display, storage, etc.
 - If a problem is found, it shows an error code.
- Boot Device Selection**
 - BIOS/UEFI checks the boot order (HDD, SSD, USB, CD/DVD).
 - Looks for a bootable device.
- Boot Loader Execution**
 - Reads the boot sector or EFI partition.
 - Loads the boot loader program into memory.
- Operating System Loading**
 - Boot loader loads the OS kernel and essential files.
 - Device drivers are initialized.
- System Ready**
 - Operating system starts completely.
 - User login screen or desktop is displayed.

5) Diagram of Booting Process



6) Role of Important Components

Component	Role in Booting
Power Supply (SMPS)	Supplies power to all hardware components.
BIOS/UEFI	Initializes hardware and starts the boot process.
POST	Checks hardware for errors.
Boot Device	Provides the boot sector/boot files.
Boot Loader	Loads the OS kernel into memory.
Operating System	Loads drivers and starts system services.
RAM	Temporary memory where OS is loaded.
Hard Disk / SSD	Stores OS files and boot loader.

7) Booting from Different Devices

- Hard Disk / SSD**: Most common boot device, contains the installed OS.
- USB Flash Drive**: Used for installing OS or booting a live OS (e.g., Linux).
- CD/DVD**: Used to install OS from disc.
- Network Booting**: System boots from a server over the network (PXE boot).

8) BIOS vs UEFI Booting

BIOS	UEFI
Older firmware	Modern firmware
Uses MBR partition table	Uses GPT partition table
Supports storage up to 2 TB	Supports very large storage
Slower boot process	Faster boot process
Text based interface	Graphical interface
Limited features	More secure and advanced features (Secure Boot)

9) Troubleshooting Boot Problems

- No power supply (check SMPS and cable).
- BIOS not detecting hard disk (check connections).
- POST errors (RAM, keyboard, display issues).
- Invalid boot device (change boot order).
- Operating system not loading (repair boot loader).
- Blue screen or error during boot (driver or OS file problem).

10) Key Points to Remember

- Booting loads the OS into main memory.
- BIOS/UEFI and POST are the first steps.
- Boot loader helps in loading the OS.
- Booting can be cold, warm or soft.
- Boot device can be HDD, SSD, USB, CD/DVD or network.
- UEFI is faster and more advanced than BIOS.
- Without a successful boot process, the system cannot be used.

"Booting is the First Step Towards a Working Computer!"